

# Revisiting and Extending Q-MAML for High-Energy Physics: Classification, Verified VQE Baselines, and Genuine Lattice Hamiltonians

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## Abstract

We re-test Q-MAML [1] (AAAI 2025), a meta-learned PQC initializer, on HEP problems it was never evaluated on. First, we audit the ML4Sci GSoC 2025 codebase that applies it to HEP classification and find both attempts broken: in one, the training loss is detached from the computation graph before backpropagation, so the learner network never actually updates; the other runs a first-order-MAML-style inner loop, not the paper’s published algorithm. We build a paper-faithful re-implementation and calibrate it against the paper’s own VQE settings at its published 2000-iteration budget (table below). Applied to HIGGS classification, a novel setting for this algorithm, it gets the best validation accuracy at every qubit count tested, significant vs. Gaussian init and the GSoC baseline ( $p < 0.05$ ,  $n = 4$  seeds). A McClean-et-al. barren-plateau scan shows no scheme, including ours, changes the variance-scaling exponent; classification cost functions plateau  $\sim 8$  orders of magnitude faster than the VQE costs the paper was validated on. Finally we run ten genuine lattice/field-theory (and adjacent) Hamiltonian pilots (Schwinger model,  $\mathbb{Z}_2$  lattice gauge theory,  $\phi^4$  scalar field,  $SU(2)$  lattice gauge theory, collective neutrino oscillations, supersymmetric quantum mechanics, non-Abelian  $\mathbb{D}_8$  lattice gauge theory, light nuclei in lattice pionless EFT, scalar Yukawa coupling, and the Lipkin-Meshkov-Glick model), now  $n=3$ -seed replicated for all ten: four robust wins, four statistical ties (two of which looked like clean single-seed wins until replicated – direct evidence for why we replicate), a seed-robust null result (too small a system to differentiate any scheme, and still null at a larger system size), and one diagnosed loss. **The diagnosed loss turns out to be an ansatz artifact, not a Q-MAML weakness:** switching to a structurally gauge-invariant ansatz (vs. the headline penalty-enforced one) makes Q-MAML the best scheme at all three seeds,  $3\times$ – $10\times$  ahead of the best classical scheme. A second qubit-count axis, tried on two tracks, gives different outcomes: one track’s machine-precision win at its smallest size gives way at 8 qubits to a seed-unstable Q-MAML (best at one of three seeds, catastrophic at another) while a classical scheme becomes the reliable choice, whereas another track’s win is more decisive at 6 qubits than at 4 – Q-MAML’s advantage is seed-, scale-, and ansatz-dependent, not fixed. A quantum-Fisher-information probe gives the paper’s “landscape placement, not plateau resistance” explanation quantitative content: Q-MAML’s predicted initializations sit at consistently higher effective dimension than classical schemes at every qubit count tested. A second classification dataset (quark-gluon jet tagging, 2 seeds) replicates the HIGGS win directionally ( $\sim 18$ -point accuracy gap, not yet significance-tested at this seed count).

| Setting (classical: 1500–2000 iters. to converge) | Speedup $S$    |
|---------------------------------------------------|----------------|
| Heisenberg XYZ (reconstructed ansatz)             | 4–6 $\times$   |
| H2 molecule (paper’s exact ansatz)                | 15–20 $\times$ |

# 1 Introduction

Variational quantum algorithms for HEP are bottlenecked by barren plateaus and by the cost of each circuit evaluation, which makes fast convergence matter more than in classical optimization. Q-MAML [1] (arXiv:2501.05906) trains a classical “Learner” network to predict PQC initial parameters via a meta-objective backpropagated with no inner-loop adaptation (Algorithm 1). The paper validates this on two VQE settings only: Heisenberg XYZ and molecular Hamiltonians.

Two gaps follow. Q-MAML has never been tested on HEP classification, the explicit motivation of the 2025 GSoC project this work builds on: auditing that codebase (§3), we find neither of its two attempts actually runs the published algorithm. And it has never been tested on a Hamiltonian that is itself particle physics: lattice field theories, which is what the 2026 proposal (QMLHEP8) motivating this work asks for. §3–6.6 close the first gap, §7 the second.

## Contributions.

1. Audit of the GSoC 2025 codebase: a detached-gradient bug and an algorithmic divergence from Algorithm 1 (§3).
2. A paper-faithful Algorithm 1/2 implementation, validated against the paper’s own two VQE settings before extension to classification (§4, §6.2–6.3).
3. Q-MAML vs. classical inits (zero,  $\pi$ , reduced-domain uniform, Gaussian) vs. the original GSoC baseline, across qubit counts, with gradient-norm diagnostics (§6.1).
4. A McClean-et-al. barren-plateau variance-scaling scan in both the VQE and classification domains (§6.4–6.6).
5. A pilot on ten lattice/field-theory (and adjacent) Hamiltonians (Schwinger,  $\mathbb{Z}_2$  LGT,  $\phi^4$  scalar field, SU(2) LGT, collective neutrino oscillations, SUSY QM,  $\mathbb{D}_8$  LGT, light nuclei in lattice pionless EFT, scalar Yukawa coupling, LMG), what the GSoC 2026 proposal asks for and untested by the original paper or contributions 1–4, with several findings that revise claims made earlier in this paper (§7).

## 2 Related Work

## 3 Auditing the Prior Codebase

Two independent GSoC 2025 attempts to apply Q-MAML to HEP classification both deviate from the published algorithm, in different ways:

- **Quark-gluon pipeline:** runs several steps of gradient descent on the PQC parameters before computing the outer (meta) loss, then stops gradient flow through the adapted parameters. Writing  $\theta_0$  for the Learner’s prediction and  $\theta_T$  for the parameters after  $T$  inner steps, the forward pass uses  $\theta_T$  but the backward pass sees  $\partial\theta_T/\partial\theta_0 \equiv \mathbb{K}$  (stop-gradient), i.e. a first-order-MAML-style variant, not the paper’s inner-loop-free Algorithm 1.
- **Higgs pipeline:** the per-task loss is converted to a plain (non-differentiable) scalar before being accumulated and only re-wrapped as a fresh, disconnected tensor at the end for back-propagation, so the gradient the Learner receives carries no information about the training data. The Learner’s weights never update across the reported 10 training epochs. A companion notebook is gradient-correct but only ever trains on one constant task embedding, which is not meta-learning in any meaningful sense.
- **Engineering:** missing package-init files broke the top-level entry point’s own imports as committed; a pinned third-party model snapshot was incompatible with current library releases;

| Area                     | Refs     | Relation to this work                                                                                                                                                                                                                                                                                                       |
|--------------------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Barren plateaus          | [4–8]    | McClean et al. variance-decay result and follow-ons: noise-induced plateaus, expressibility-gradient link, Lie-algebraic onset theory, cost-function-dependent plateaus (the last explains §6.5).                                                                                                                           |
| Init. baselines          | [10, 11] | Gaussian and reduced-domain-uniform init, our classical comparison points.                                                                                                                                                                                                                                                  |
| Meta-learning            | [2, 3]   | MAML (classical); Meta-VQE (task-conditioned energy profiles, predates Q-MAML).                                                                                                                                                                                                                                             |
| Q-MAML                   | [1]      | AAAI 2025, arXiv:2501.05906. Validated only on VQE (Heisenberg, molecules).                                                                                                                                                                                                                                                 |
| Concurrent init. work    | [12–14]  | Qracle (GNN param. predictor, no cross-task meta-objective); DiffQ (diffusion-model param. sampling, no MAML-style adaptation); “Qubits to Couplings” (per-event hypernetwork for $HH \rightarrow b\bar{b}\gamma\gamma$ , doesn’t cite Finn et al. or Q-MAML). None test Lee/Cho/Kim’s Algorithm 1/2 on HEP classification. |
| Genuine HEP Hamiltonians | [16–19]  | Trapped-ion Schwinger-model VQE; barren-plateau-free $\mathbb{Z}_2$ LGT (uses a Gauss-law warm start, no learned init, our Track B comparison point); scalar-field qubit encoding we build Track C from; QCNN phase classification on pre-prepared states (different mechanism, no overlap with Track A).                   |

Table 1: Related work by area. No prior work applies a Learner-predicts-init mechanism to a lattice field theory Hamiltonian; §7 is, to our knowledge, the first.

a configuration module created a directory named after its own placeholder path string on import.

## 4 Method

### 4.1 Paper-faithful Q-MAML (Algorithm 1/2)

Per task  $T_i$  with cost  $l_{T_i}(\theta) = \langle \psi(\theta) | \hat{H} | \psi(\theta) \rangle$ , a Learner  $h_W$  maps task descriptor  $\phi_i \rightarrow \theta = h_W(\phi_i)$ , trained via (paper’s Eq. 2)

$$\arg \min_W \sum_{T_i \sim p(T)} l_{T_i}(g(h_W(\phi_i))), \quad (1)$$

backpropagated with no inner-loop adaptation (Eq. 3: the difference from true MAML). We keep Eq. 1 exactly and swap the domain-specific pieces (Table 2).

Pretraining (classification and VQE) shares this identical structure, differing only in the cost function. Evaluation (Algorithm 2): freeze the Learner, set  $\theta_0 = h_{W^*}(\phi^*)$  for a held-out task, run gradient descent on  $l_{T^*}(\theta)$ . This contrasts with the GSoC baseline (§3); both are reported, neither treated as a straw man.

Table 2: VQE setting (paper)  $\rightarrow$  classification setting (this work).

| Paper (VQE)                                               | This work (classification)                                                                                                                                            |
|-----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Task = Hamiltonian instance, e.g. $J^i = [J_x, J_y, J_z]$ | Task = quantile bin of a physics variable (reconstructed Higgs-candidate mass for HIGGS; jet transverse momentum for quark-gluon), class-balanced support/query split |
| Descriptor $\phi_i = J^i$                                 | mean encoder embedding of the support set                                                                                                                             |
| $l_{T_i}(\theta) = \langle \psi   \hat{H}   \psi \rangle$ | $l_{T_i}(\theta) = \text{CE}(\text{PQC}_\theta(\text{support}_i), y_i)$                                                                                               |
| PQC $g(\theta)$                                           | Same hardware-efficient ansatz family + classical encoder producing the encoding angles                                                                               |

## 4.2 Architecture

Classical encoder (a convolutional backbone for jet images; a small MLP for tabular HIGGS features)  $\rightarrow$  PQC angles  $\rightarrow$  hardware-efficient ansatz (single-qubit rotations plus a ring of entangling gates), evaluated via state-vector simulation with adjoint differentiation where supported (parameter-shift otherwise)  $\rightarrow$  linear classification head. Learner: a 2-layer MLP mapping the task embedding to the PQC weight tensor.

## 4.3 Baselines

Zero,  $\pi$ , reduced-domain uniform [11], Gaussian [10] init (per-task adaptation with no Learner); the GSoC baseline (§3).

## 4.4 Metrics, formally

For a held-out task with exact ground energy  $E_0$  (sparse diagonalization) and PQC energy  $E_{\text{PQC}}(\theta_t)$  at adaptation step  $t$ , we report a log-gap and a final relative error:

$$\Delta(t) = \log(|E_{\text{PQC}}(\theta_t) - E_0| + \epsilon), \quad \epsilon = 10^{-8}, \quad \varepsilon_{\text{rel}} = \frac{|E_{\text{PQC}}(\theta_T) - E_0|}{\langle |E_0| \rangle_{\text{test tasks}}}. \quad (2)$$

Convergence speedup of Q-MAML over a classical scheme  $A$ , at a fixed error threshold  $\Delta^*$ :

$$S = \frac{t_A(\Delta < \Delta^*)}{t_{\text{Q-MAML}}(\Delta < \Delta^*)}. \quad (3)$$

Classification significance: Welch’s  $t$ -test and Cohen’s  $d = (\bar{x}_1 - \bar{x}_2)/s_p$  with pooled standard deviation  $s_p = \sqrt{(s_1^2 + s_2^2)/2}$ ,  $n = 4$  seeds per arm. Barren-plateau scan:  $\text{Var}_{\theta, \text{task}}[\partial C / \partial \theta_{\text{ref}}]$  at a fixed reference parameter, sampled across many random inits/tasks, vs. qubit count at fixed depth.

## 5 Experimental Setup

- **Data:** HIGGS (UCI, 180k-row streamed subsample, primary); Quark-Gluon (ML4Sci, access requested, pending).
- **Tasks:** quantile-bin the Higgs-candidate mass into 6 bins; support=8/query=8 per bin; 3 tasks/bin train (18 total), 2/bin test (12 total).
- **Qubits:** 4/6/8 (classification); 4–14 (VQE-domain barren-plateau scan).

- **Depth:** 2 (classification), 3/6 (VQE barren-plateau scan), 3 (Heisenberg), 7 (H2, matches paper).
- **Seeds:** 4 (classification headline).
- **Compute:** CPU only, state-vector simulation with adjoint differentiation (parameter-shift fallback).
- **Hyperparameters:** 5 inner-loop steps at inner learning rate 0.02, outer learning rate  $5 \times 10^{-3}$ , initial-weight scale 0.01 (classification); VQE pretraining learning rate 0.001 over 30 epochs (matches paper); VQE adaptation learning rate 0.001 over 2000 iterations (matches paper).

## 6 Results

### 6.1 HEP classification

HIGGS, 10 epochs, 4 seeds, 4/6/8 qubits.

Table 3: Final validation accuracy (mean  $\pm$  std,  $n = 4$ ).

| Qubits | Q-MAML                              | Zero              | Uniform           | Gaussian          | GSoC baseline     |
|--------|-------------------------------------|-------------------|-------------------|-------------------|-------------------|
| 4      | <b>0.589 <math>\pm</math> 0.035</b> | 0.534 $\pm$ 0.037 | 0.510 $\pm$ 0.019 | 0.487 $\pm$ 0.034 | 0.505 $\pm$ 0.049 |
| 6      | <b>0.576 <math>\pm</math> 0.033</b> | 0.497 $\pm$ 0.031 | 0.513 $\pm$ 0.060 | 0.503 $\pm$ 0.032 | 0.523 $\pm$ 0.043 |
| 8      | <b>0.563 <math>\pm</math> 0.023</b> | 0.513 $\pm$ 0.049 | 0.523 $\pm$ 0.059 | 0.518 $\pm$ 0.023 | 0.503 $\pm$ 0.036 |

Table 4: Q-MAML vs. baselines: Welch’s  $t$ -test  $p$  / Cohen’s  $d$ .

| Qubits | vs. Gaussian         | vs. GSoC baseline    | vs. best classical (post-hoc) |  |  |
|--------|----------------------|----------------------|-------------------------------|--|--|
| 4      | $p=0.006$ , $d=2.95$ | $p=0.035$ , $d=1.98$ | $p=0.076$ , $d=1.52$          |  |  |
| 6      | $p=0.020$ , $d=2.22$ | $p=0.106$ , $d=1.36$ | $p=0.134$ , $d=1.28$          |  |  |
| 8      | $p=0.034$ , $d=1.94$ | $p=0.035$ , $d=2.02$ | $p=0.287$ , $d=0.87$          |  |  |

| Meta-loss          |  | 4 qubits         | 6 qubits | 8 qubits |
|--------------------|--|------------------|----------|----------|
| Q-MAML             |  | 0.382            | 0.360    | 0.325    |
| Every other scheme |  | 0.68–0.70 (flat) |          |          |

Significant against Gaussian and the GSoC baseline at all three qubit counts; against the post-hoc best classical scheme, directionally positive but not significant at  $n = 4$  (underpowered, not negative: the comparator is selected after the fact per qubit count). Q-MAML is the only variant whose training loss moves (Figure 1).

Table 5: Correction: accuracy gap (Q-MAML – best classical) at 2 vs. 4 seeds.

| Qubits             | 4      | 6      | 8      |
|--------------------|--------|--------|--------|
| Gap at $n=4$ seeds | +0.055 | +0.063 | +0.039 |

**Correction.** An earlier 2-seed pass showed the gap reversing (going negative) by 8 qubits. At  $n = 4$  seeds (Table 5) it does not: Q-MAML stays ahead at every qubit count, with a modest, non-monotonic decline directionally consistent with §6.5 but not a confirmed match.

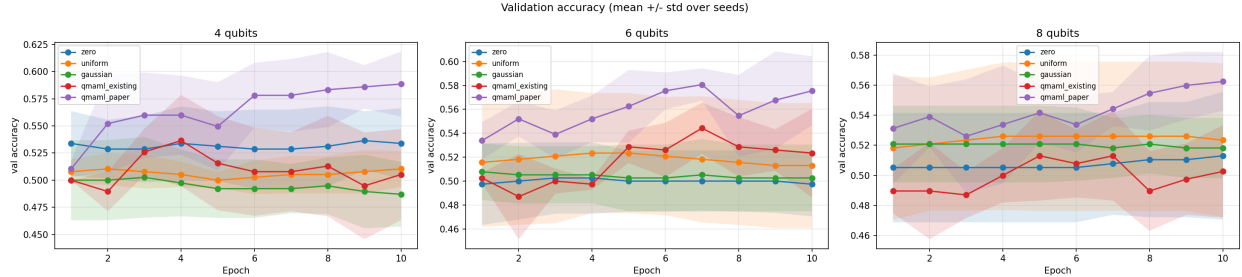


Figure 1: Validation accuracy vs. epoch, 4/6/8 qubits, mean  $\pm$  std over 4 seeds.

Two other findings. Q-MAML’s validation loss is worse (0.80–0.87, worsening with qubit count) than every other scheme ( $\sim 0.69$ ) despite best accuracy: a calibration gap, not a clean sweep. Its mean gradient norm (1.36–1.61) is higher than classical schemes (0.27–0.64), the opposite of the paper’s “moderate norm is better” framing: here higher norm tracks with best accuracy at every qubit count.

**Quark-gluon (second dataset, generalization check).** 6 qubits, 10 epochs, 2 seeds, jet-substructure features. Q-MAML again beats every classical scheme by a wide margin: validation accuracy  $0.776 \pm 0.005$  vs. 0.594–0.599 (zero/uniform/Gaussian, tightly clustered); meta-loss  $0.308 \pm 0.003$  vs. 0.688–0.701 (near chance level,  $\ln 2 \approx 0.693$ , for every classical scheme). No ‘zero’-specific anomaly here, matching HIGGS and unlike the VQE-domain fixed-point pattern. Only 2 seeds (effect size  $\sim 6.8$  in raw terms, not a powered test): read as directional confirmation the HIGGS result generalizes, not a sharper result than HIGGS’s own 4-seed numbers.

## 6.2 Molecule Hamiltonian VQE

H2, 4 qubits, depth 7, 2000 iterations, paper’s exact ansatz.

Table 6: H2 molecule: iteration to reach the reported quality, and final energy (2000 iterations).

| Scheme                  | Iteration to converge                                | Final energy         |
|-------------------------|------------------------------------------------------|----------------------|
| Q-MAML                  | $\sim 100$ (machine precision, $\Delta \sim -17.3$ ) | $-17.30$             |
| $\pi$ /uniform/Gaussian | $\sim 1500$ –2000                                    | $-17.04$ to $-17.30$ |
| Zero                    | fixed at start value                                 | (unchanged)          |

By iteration 2000 all four non-degenerate schemes converge to the same final energy (Table 6): the advantage is speed ( $S \approx 15$ –20, Eq. 3), not a different final optimum, matching §6.3. This uses the paper’s exact circuit, so it is the strongest evidence our system implements Algorithm 1/2 correctly.

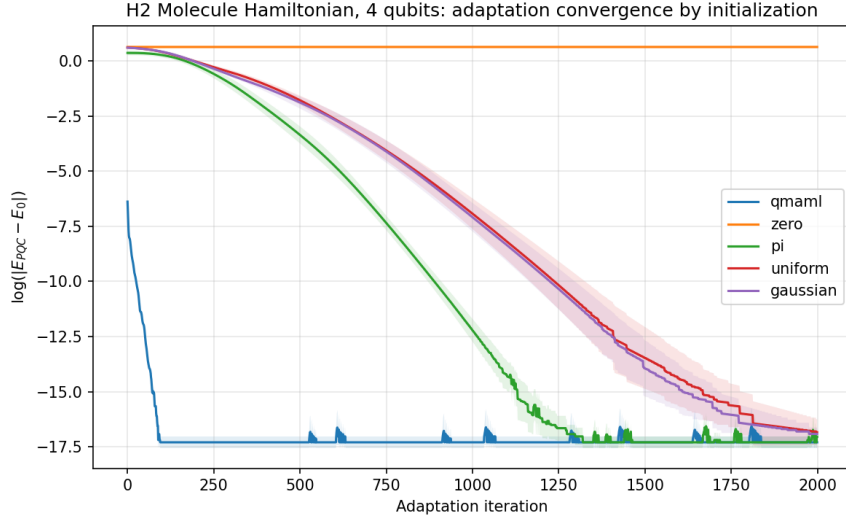


Figure 2: H2 molecule Hamiltonian: adaptation convergence by init scheme, 2000 iterations.

### 6.3 Heisenberg XYZ VQE

6 qubits, depth 3, 8 held-out tasks, 2000 iterations (paper’s budget, reconstructed ansatz: paper’s appendix circuit unavailable to us).

Table 7: Heisenberg XYZ: log-gap  $\Delta$  at iteration 0 and at iteration 2000.

|                | Q-MAML     | Uniform    | Gaussian   | $\pi$      | Zero       |
|----------------|------------|------------|------------|------------|------------|
| $\Delta(0)$    | $\sim 1.5$ | $\sim 2.5$ | $\sim 2.5$ | $\sim 2.5$ | $\sim 2.5$ |
| $\Delta(2000)$ | 0.711      | 0.736      | 0.727      | 0.966      | 2.522      |

At 300 iterations (earlier pass) Q-MAML held a large lead (0.82 vs. 1.71–1.76); at 2000 iterations that gap mostly closes (Table 7). Q-MAML flattens by iteration  $\sim 250$ –500 while uniform/Gaussian need  $\sim 1500$ –2000,  $S \approx 4$ –6. Q-MAML starts closer to the ground state before any adaptation, the initialization-quality claim the paper actually makes; the speed result follows from it.

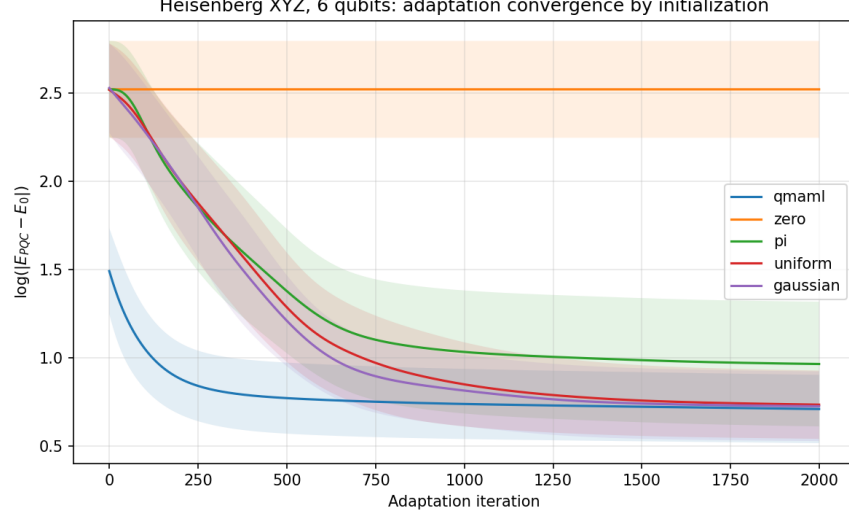


Figure 3: Heisenberg XYZ, 6 qubits: adaptation convergence by init scheme, 2000 iterations.

#### 6.4 Barren-plateau scaling, VQE domain

Qubits 4–14, depths  $\{3, 6\}$ , 80 samples per point. Zero/ $\pi$  (constant) init show zero variance trivially (no sample-to-sample variation, not plateau resistance). Among the random schemes (Q-MAML, uniform, Gaussian): no exponential decay over 4–10 qubits at depth 3, likely too shallow/small for the asymptotic regime. Q-MAML’s variance is comparable to or larger than classical schemes at every qubit count: no evidence it changes the scaling exponent; its VQE advantage (§6.3) is landscape placement, not plateau resistance.

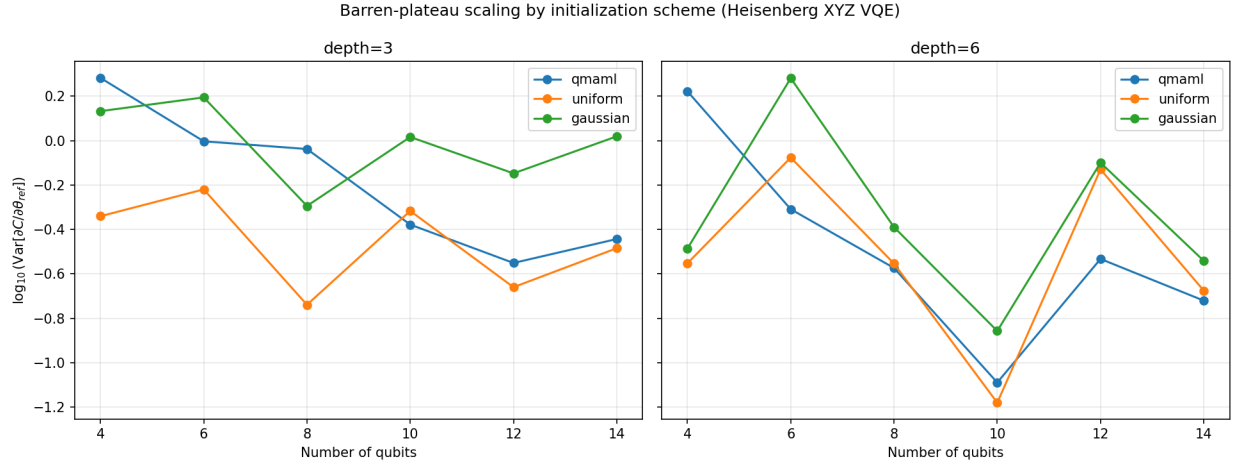


Figure 4: Barren-plateau variance scaling by init scheme, VQE domain (depths 3 and 6).

**A second, independent diagnostic gives “landscape placement” a quantitative meaning.** The scan above uses a single reference-parameter gradient per sample (the standard McClean et al. protocol), which cannot distinguish “degenerate everywhere” from “degenerate along one direction, rich along others.” We additionally compute the quantum Fisher information matrix (via PennyLane’s `metric_tensor`, block-diagonal approximation) at each sampled parameter point and summarize it as a participation ratio  $PR = (\sum \lambda)^2 / \sum \lambda^2$  normalized by parameter count – an effective dimension: how many directions in parameter space the circuit can actually distinguish, independent of and complementary to a single-coordinate gradient. Qubits {4, 6, 8}, depth 3, Heisenberg XYZ ansatz, 8 samples per (qubit count, scheme).

Table 8: Effective dimension (participation ratio / # parameters), Heisenberg XYZ ansatz.

| Qubits | Q-MAML                              | Uniform           | Gaussian          |
|--------|-------------------------------------|-------------------|-------------------|
| 4      | $0.933 \pm 0.011$                   | $0.683 \pm 0.011$ | $0.715 \pm 0.033$ |
| 6      | $0.934 \pm 0.025$                   | $0.687 \pm 0.011$ | $0.726 \pm 0.019$ |
| 8      | <b><math>0.958 \pm 0.016</math></b> | $0.693 \pm 0.011$ | $0.733 \pm 0.018$ |

Q-MAML’s predicted initializations sit at consistently higher effective dimension (93–96% of the maximum possible rank) than Uniform (68–69%) or Gaussian (72–73%) at every qubit count tested, and the gap does not shrink with qubit count – if anything Q-MAML’s edge widens slightly at 8 qubits. This is consistent with, and gives quantitative content to, the “landscape placement” explanation above: by this independent metric, Q-MAML’s initialization is not merely comparable in gradient magnitude but sits in a genuinely less-degenerate region of parameter space than random classical schemes, a distinction the single-reference-gradient scan alone cannot make. Reduced qubit range (4–8 vs. the gradient scan’s 4–14) and sample count (8 vs. 80 per point) for tractability – `metric_tensor` is markedly more expensive per sample than a single gradient component; extending this to the full range and to the classification domain is future work.

## 6.5 Barren-plateau scaling, classification domain

HIGGS.

All three schemes (Q-MAML, uniform, Gaussian) show  $\sim 8$  orders of magnitude exponential decay from 4 to 10 qubits: a textbook plateau, unlike the flat VQE trend at the same qubit range, consistent with classification cross-entropy behaving as a “global” cost function [5]. By 10 qubits every scheme collapses to the same floor,  $\sim 10^{-35}$ .

Table 9: Gradient variance at 6 qubits: pretrain-length sensitivity.

| Pretrain length  | Q-MAML                | Uniform               | Gaussian              |
|------------------|-----------------------|-----------------------|-----------------------|
| Short (8 epochs) | at/below classical    | baseline              | baseline              |
| Long (25 epochs) | $1.65 \times 10^{-6}$ | $4.65 \times 10^{-8}$ | $3.93 \times 10^{-7}$ |

Below the collapse point, ranking is unstable (Table 9): a pretrain-length artifact, not a stable ranking. No scheme dominates at any qubit count. The 8–10-qubit collapse-point range itself is unchanged.

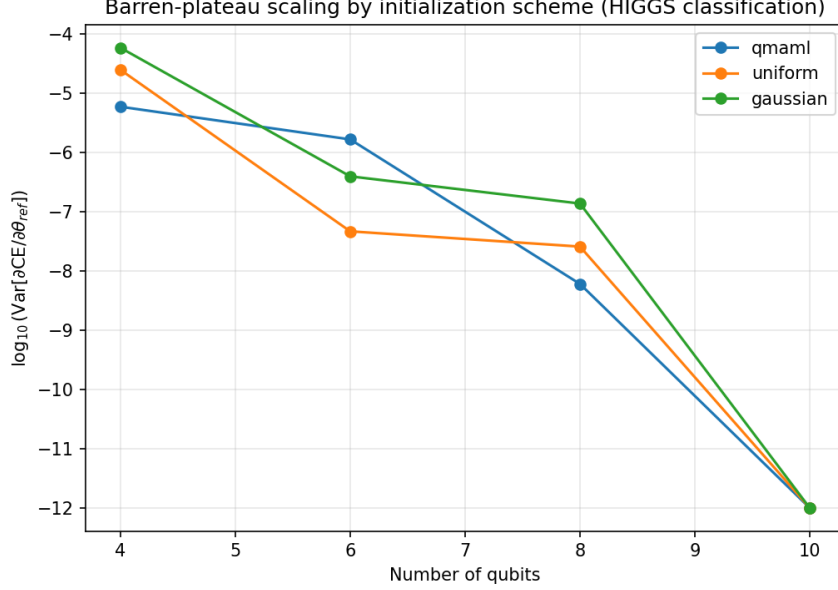


Figure 5: Barren-plateau variance scaling by init scheme, HIGGS classification domain.

## 6.6 Combined takeaway

(1) Q-MAML’s benefit is landscape placement, not a change in the barren-plateau scaling exponent, in either domain. (2) HEP classification plateaus faster than VQE for every scheme tested: a challenge for the QMLHEP direction generally, not specific to Q-MAML.

## 7 Q-MAML on Genuine HEP Hamiltonians: A GSoC 2026 Pilot

Sections above test Q-MAML on HEP *data* (HIGGS) or the paper’s own non-HEP VQE settings. Neither is what QMLHEP8 asks for: a Learner initializing VQEs for Hamiltonians that are themselves lattice field theories. Table 10 summarizes the ten tracks piloted here (2–8 qubits, verified for correctness before use, metrics of §4.4). Every track except H is now  $n=3$ -seed replicated (Tracks B/C were already multi-seed at specific points; A/D/E/F/G/I/J are newly replicated here), and Tracks D and E additionally have one larger-qubit-count extension (single seed) verified and run.

Table 10: Track overview.

| Track                 | Qubits          | Structure                                 | Q-MAML verdict                                                                                                |
|-----------------------|-----------------|-------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| A: Schwinger          | 4–8             | U(1) gauge eliminated analytically        | Wins, depth-dep. ( $n=3$ at $L=2$ )                                                                           |
| B: $\mathbb{Z}_2$ LGT | 8               | Gauge symmetry, penalty-enforced          | Worst on headline ansatz ( $n=3$ ); best at 3/3 seeds on gauge-invariant ansatz                               |
| C: $\phi^4$ scalar    | 6               | Bosonic, no gauge/fermion structure       | Wins (interp. only, $n=3$ )                                                                                   |
| D: SU(2) LGT          | 4 (+6, +8 ext.) | Non-Abelian gauge eliminated analytically | Clean win at $N=2$ ; at $N=3$ $\pi$ best, Q-MAML second; seed-unstable at $N=4$ ( $\pi$ robust) ( $n=3$ each) |
| E: neutrino           | 4 (+6 ext.)     | All-to-all, no locality, reformulated     | Wins 2/3 seeds at $N=4$ ; clean win at $N=6$                                                                  |
| F: SUSY QM            | 4               | Discrete + continuous task space          | Ties Uniform/Gaussian ( $n=3$ ); beats hand-picked init                                                       |
| G: $\mathbb{D}_8$ LGT | 7               | Non-Abelian, un-reduced, fermionic matter | Ties Uniform ( $n=3$ ); beats $\pi$ /Zero                                                                     |
| H: nuclear EFT        | 8               | UCCSD ansatz, 3-nucleus discrete axis     | Ties (same final, faster; $n=3$ , strongest confirmation)                                                     |
| I: Yukawa             | 4               | Fermion-boson, reformulated               | Ties with zero-init ( $n=3$ , confirmed)                                                                      |
| J: LMG                | 2               | Nuclear-structure-adjacent, not HEP       | Null result ( $n=3$ , confirmed)                                                                              |

## 7.1 Track A: lattice Schwinger model

Single-flavor Kogut-Susskind staggered fermions, gauge field eliminated via Gauss’s law; task params  $(m_0, g)$ . Hardware-efficient ansatz, 16 train/6 test tasks, 40 pretrain epochs, 300 adaptation iterations.

Table 11:  $L=2$  (4 qubits, depth 3): final relative error  $\varepsilon_{\text{rel}}$  and iteration to reach  $< 1\%$  error.

|                            | Q-MAML     | $\pi$ | Gaussian | Uniform | Zero  |
|----------------------------|------------|-------|----------|---------|-------|
| $\varepsilon_{\text{rel}}$ | $\sim 0\%$ | 0.06% | 0.19%    | 1.61%   | 396%  |
| Iteration $< 1\%$          | 4          | 192   | 266      | never   | never |

Q-MAML converges fastest and most accurately at  $L=2$  (Table 11). Its init log-gap ( $-3.29$ ) is already better than any classical scheme’s starting point. Zero-init is a fixed point (exactly flat error) for the whole run.

At  $L=3$  (6 qubits), same depth,  $\pi$  overtakes Q-MAML by the final iteration (32.95% vs. 38.08%, Table 12). Pretrain gradient norm stays healthy: points to insufficient ansatz depth, not a plateau.

Depth 6/9 recovers and then extends Q-MAML’s lead ( $\sim 10\times$  better than  $\pi$  at depth 9): the  $L=2 \rightarrow L=3$  dip is a depth-insufficiency artifact of reusing the  $L=2$  depth, not a breakdown at larger qubit count. Zero-init stays at an identical error at every depth (algebraic fixed point, depth-independent).

The depth story does not extrapolate cleanly to  $L=4$ : extending “depth  $\approx 1.5\times$  qubits” to depth 12, Q-MAML loses to  $\pi$  (Table 13). Unlike the  $L=3$  case, pretraining itself fails to converge, not

Table 12: Final relative energy error  $\varepsilon_{\text{rel}}$  vs. ansatz depth,  $L = 3$  (6 qubits).

| Depth | Q-MAML       | $\pi$         | Gaussian | Uniform | Zero    |
|-------|--------------|---------------|----------|---------|---------|
| 3     | 38.08%       | <b>32.95%</b> | 46.24%   | 56.12%  | 697.64% |
| 6     | <b>2.63%</b> | 3.06%         | 12.92%   | 9.40%   | 697.64% |
| 9     | <b>0.05%</b> | 0.57%         | 1.42%    | 1.35%   | 697.64% |

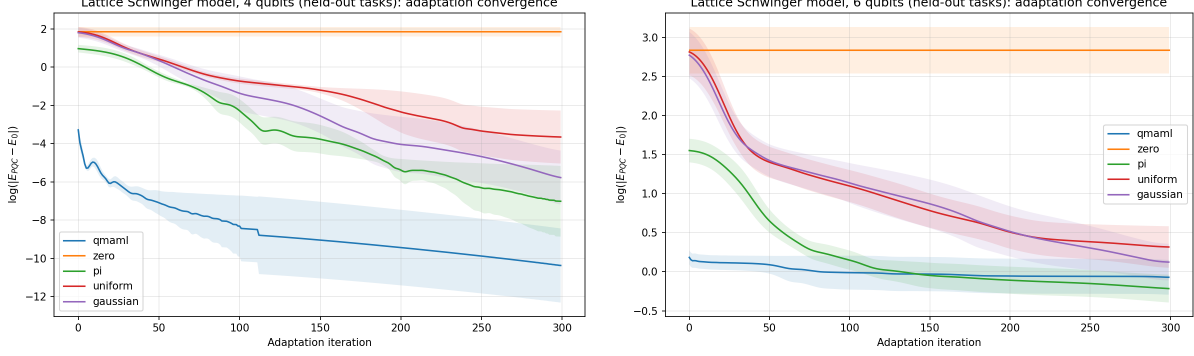


Figure 6: Schwinger model adaptation convergence:  $L = 2/\text{depth } 3$  (left) vs.  $L = 3/\text{depth } 3$  (right).

just the ansatz. Open question: whether pretrain budget, not just depth, needs to scale with qubit count. One spot-check, not a scan.

**The  $L=4$  failure has two causes, and fixing both recovers Q-MAML’s lead.** A pretrain-budget scan at  $L=4$  (8 qubits; single seed, reduced protocol: 3 held-out tasks, 150 adaptation iterations, so absolute numbers are not comparable to Table 13, but the 40-epoch row reproduces its pretraining signature exactly: mean energy 7.21, gradient norm 24.6):

Table 14:  $L=4$  depth  $\times$  pretrain-budget scan: final relative error, and pretrain diagnostics.

| Depth, epochs, lr | Q-MAML       | $\pi$        | Uniform | Gaussian | pretrain $E$ | pretrain $\ g\ $ |
|-------------------|--------------|--------------|---------|----------|--------------|------------------|
| 12, 40, 0.005     | 24.5%        | <b>17.9%</b> | 39.4%   | 38.4%    | 7.21         | 24.6             |
| 12, 100, 0.005    | 23.9%        | <b>17.9%</b> | 39.4%   | 38.4%    | 7.03         | 24.1             |
| 12, 100, 0.002    | 21.2%        | <b>17.9%</b> | 39.4%   | 38.4%    | −1.95        | 7.9              |
| 18, 100, 0.002    | <b>2.43%</b> | 8.65%        | 11.0%   | 11.8%    | −3.27        | 4.7              |

More pretraining epochs alone do nothing (24.5  $\rightarrow$  23.9%; pretraining stays stalled). Lowering the pretrain learning rate from 0.005 to 0.002 fixes pretraining itself (mean energy −1.95, inside the −2 to −4 range of the true ground energies; gradient norm 24  $\rightarrow$  7.9) but only modestly improves the adapted result (21.2%) – still behind  $\pi$ . Raising depth to 18 at the same budget then gives Q-MAML a 3.6 $\times$  lead over  $\pi$  (2.43% vs. 8.65%). So at  $L=4$  the learning rate was a genuine cause of the pretraining failure, and depth 12 was independently too shallow for 8 qubits, extending the  $L=3$  depth-insufficiency finding above one size up rather than contradicting it. Caveats: the depth effect is isolated only at  $\text{lr} = 0.002$  (depth 12 vs. 18, same epochs); depth 18 at  $\text{lr} = 0.005$  was not run, so it is untested whether the original learning rate would still have failed at the larger depth; single seed; reduced protocol.

Table 13:  $L=4$  (8 qubits, depth 12) and pretrain diagnostics.

|                                                                                | Q-MAML | $\pi$        |
|--------------------------------------------------------------------------------|--------|--------------|
| Final relative error                                                           | 31.81% | <b>8.56%</b> |
| Pretrain gradient norm: 23.6 $\rightarrow$ 24.6 over 40 epochs (no decrease)   |        |              |
| Pretrain mean energy: stalls at 7.21 (true ground energies $\sim -2$ to $-4$ ) |        |              |

## 7.2 Track B: single-plaquette $\mathbb{Z}_2$ lattice gauge theory

One plaquette of the two-leg-ladder geometry in [17] (8 qubits: 4 matter + 4 link), task params  $(J, m, \mu)$ . Gauss’s law enforced by an energy penalty

$$H_{\text{penalty}} = V \sum_l (G_l - q_l)^2, \quad V = 10, \quad (4)$$

not eliminated analytically: verified exactly ( $[H, G_l] = 0$  and  $G_l^2 = \mathbb{I}$  to machine precision).

Table 15: Track B headline: final relative error and diagnosis at  $V=10$ .

|                            | Uniform       | $\pi$  | Gaussian | Q-MAML | Zero    |
|----------------------------|---------------|--------|----------|--------|---------|
| $\varepsilon_{\text{rel}}$ | <b>23.07%</b> | 24.69% | 24.05%   | 59.16% | 112.27% |

Q-MAML’s log-gap is flat for the full 300-iteration budget ( $3.800 \rightarrow 3.791$ ) while classical schemes, starting worse ( $\Delta(0) \sim 4.43$ ), descend substantially: the opposite of Track A.

**Diagnosis.** The gradient at Q-MAML’s predicted init is small but nonzero (5.73, vs. zero-init’s exact 0); learning rates 0.02/0.1/0.3 all plateau at the same cost within 100 iterations: a landscape plateau, not a fixed point. Leading hypothesis: at  $V=10$  the penalty term (Eq. 4,  $O(10)$ ) dominates the physical terms ( $O(1)$ ), so pretraining mostly reduces constraint violation instead of encoding  $(J, m, \mu)$ .

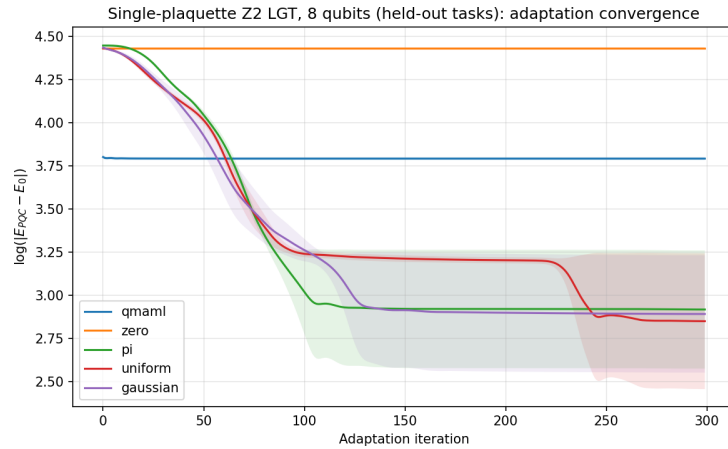


Figure 7:  $\mathbb{Z}_2$  LGT: adaptation convergence by init scheme. Q-MAML is not competitive here.

Table 16: Final relative energy error vs. penalty strength  $V$  (Eq. 4).

| $V$ | Q-MAML  | $\pi$   | Gaussian | Uniform        | Zero    |
|-----|---------|---------|----------|----------------|---------|
| 1   | 149.49% | 163.97% | 156.96%  | <b>145.08%</b> | 415.51% |
| 3   | 77.39%  | 51.75%  | 40.78%   | <b>27.68%</b>  | 148.56% |
| 10  | 59.16%  | 24.69%  | 24.05%   | <b>23.07%</b>  | 112.27% |
| 30  | 27.19%  | 17.73%  | 27.42%   | <b>16.49%</b>  | 103.91% |

The  $V$ -sweep does not confirm the penalty-dominance hypothesis: Q-MAML plateaus (flat curve) at every  $V \in \{1, 3, 10, 30\}$ , ties the best classical scheme only at  $V=1$ , and  $V=3$  (not  $V=30$ ) is its relatively worst point ( $2.8\times$  the best scheme). Weakening  $V$  does not straightforwardly fix Q-MAML. Narrower,  $V$ -independent finding: the Learner’s pretraining on this Hamiltonian/ansatz combination reliably lands in flat regions, mechanism unidentified. Most promising untried fix: a gauge-invariant, structurally (not penalty-) invariant ansatz [17] (§9).

Table 17: 3-seed replication,  $V=10$ : mean  $\pm$  std relative error (per-seed values in parentheses).

| Scheme   | Relative error, 3 seeds                    |
|----------|--------------------------------------------|
| Q-MAML   | $49.38\% \pm 13.70\%$ (59.16/30.01/58.96%) |
| Uniform  | $20.34\% \pm 3.65\%$                       |
| Gaussian | $19.43\% \pm 4.76\%$                       |
| $\pi$    | $27.35\% \pm 3.22\%$                       |
| Zero     | $111.87\% \pm 1.48\%$                      |

Q-MAML’s best seed (30.01%) is still worse than every uniform seed (15.18–23.07%): ranking holds seed-by-seed; magnitude is seed-sensitive (Table 17).

**The untried fix works.** We built and ran the two ready-made ansätze from [17] at  $V=10$ : **Z2LGTHVAPQC**, a gauge-invariant Hamiltonian-variational ansatz (each layer Trotterizes the individual Pauli terms of the *physical* Hamiltonian – hopping, staggered mass, electric, magnetic – so gauge invariance is structural, not penalty-enforced), and **Z2LGTHWEffZZPQC**, a second ready-made ansatz that is *not* gauge-invariant by construction (a same-complexity control).

Table 18: Track B, alternative ansätze at  $V=10$ , final relative error, mean  $\pm$  std over  $n=3$  seeds (reduced budget: 4 held-out tasks, 200 adaptation iterations – documented for tractability, §9).

| Ansatz                        | Q-MAML                     | Gaussian          | Uniform           | $\pi$                  | Zero             |
|-------------------------------|----------------------------|-------------------|-------------------|------------------------|------------------|
| Gauge-invariant HVA (%)       | <b>0.0022</b> $\pm$ 0.0011 | 0.012 $\pm$ 0.002 | 0.013 $\pm$ 0.007 | 0.014 $\pm$ 0.009      | 0.011 $\pm$ 0.00 |
| Hardware-eff. ZZ, control (%) | 2.58 $\pm$ 1.16            | 2.58 $\pm$ 1.16   | 2.58 $\pm$ 1.16   | <b>1.53</b> $\pm$ 0.26 | 2.68 $\pm$ 1.15  |

**On the gauge-invariant HVA, Q-MAML flips from worst scheme (Table 15) to best at every one of three seeds**,  $3\times-10\times$  ahead of the best classical scheme per seed (seed 0: 0.0031% vs. 0.0148%; seed 1: 0.0029% vs. 0.0088%; seed 2: 0.0007% vs. 0.0073%), all at  $\lesssim 0.03\%$  relative error. That is the same qualitative pattern as Track A and Track D at  $N=2$ , which share Track B’s penalty-free structural gauge invariance (Track G, also gauge-eliminated, replicated only to a tie, so the pattern is not universal across those tracks). On the non-gauge-invariant control ansatz, Q-MAML is indistinguishable from Uniform and Gaussian at every seed (identical to three significant figures within a seed), and  $\pi$  is the control’s most variable scheme (best at seeds 0 and 2, worst at seed 1) – ruling out “any different ansatz helps” as the explanation. **This isolates the mechanism:** it is gauge invariance being structural rather than penalty-enforced that fixes Track B’s Q-MAML result, confirming the hypothesis in §9 directly rather than leaving it as an untried suggestion. Caveats: the reduced 4-task/200-iteration budget (not Table 15’s full one), one system size, and the gauge-invariant and control ansätze differ in more than gauge invariance alone (they are two different published circuits, not a minimal pair), so “structural gauge invariance is the operative variable” is the best-supported reading of this comparison, not a controlled ablation.

### 7.3 Track C: $\phi^4$ lattice scalar field theory

Li/Macridin/Mrenna/Spentzouris encoding [18] (6 qubits: 2 sites  $\times$  3 qubits/site, field-amplitude truncation), task params  $(m_0^2, \lambda_0)$ . No gauge field, no fermion mapping;  $H$  is a dense Hermitian matrix (a field-amplitude-basis Hamiltonian, not a Pauli sum). Verified: conjugate-field eigenvalues match the closed-form analytic formula exactly; reconstructed harmonic-oscillator spectrum matches  $n + \frac{1}{2}$  to  $< 2\%$  on low-lying levels.

Table 19: Track C headline: final relative energy error.

|                            | Q-MAML       | Zero  | Uniform | Gaussian | $\pi$  |
|----------------------------|--------------|-------|---------|----------|--------|
| $\varepsilon_{\text{rel}}$ | <b>5.86%</b> | 5.77% | 7.11%   | 7.55%    | 36.06% |

Q-MAML reaches log-gap  $-1.74$  by iteration 50; zero/uniform/Gaussian mostly catch up by iteration 300: a speed advantage ( $S \approx 3$ ), not a different final optimum, matching §6.3/6.2.

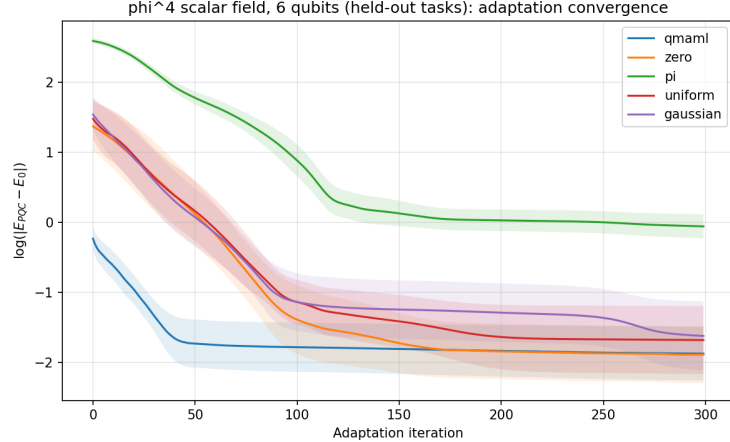


Figure 8:  $\phi^4$  scalar field: adaptation convergence by init scheme.

**Two corrections to earlier sections.** Zero-init is *not* stuck here (log-gap  $1.367 \rightarrow -1.893$ , tying Q-MAML): the only Hamiltonian in this paper where the zero-parameter fixed point doesn’t hold, so that pattern is Hamiltonian-structure-dependent.  $\pi$  (best classical scheme on Track B, 2nd-best on Track A at  $L=2$ ) is worst here (36.06%,  $\sim 5\text{--}6\times$  worse than alternatives). No classical scheme is reliable across all three tracks under interpolation; Q-MAML is.

Table 20: 3-seed replication (uniform-random split): mean  $\pm$  std relative error.

| Scheme   | Relative error, 3 seeds                 |
|----------|-----------------------------------------|
| Q-MAML   | $6.05\% \pm 1.60\%$                     |
| Zero     | $6.12\% \pm 1.54\%$                     |
| Uniform  | $6.90\% \pm 1.80\%$                     |
| Gaussian | $6.85\% \pm 1.52\%$                     |
| $\pi$    | $27.00\% \pm 9.09\%$ (best seed 14.58%) |

**Phase-stratified generalization.** This Hamiltonian’s task space spans a symmetric/broken-symmetry transition [18]. Train on symmetric ( $m_0^2 \in [0.3, 1.0]$ ), test on broken-symmetry-tending ( $m_0^2 \in [-2.5, -1.7]$ ), and reverse.

Table 21: Phase-stratified final relative energy error (single seed).

| Direction                 | Q-MAML | Zero         | Uniform       | Gaussian     | $\pi$ |
|---------------------------|--------|--------------|---------------|--------------|-------|
| Sym. $\rightarrow$ broken | 3.13%  | <b>1.81%</b> | <b>1.81%</b>  | <b>1.81%</b> | 5.04% |
| Broken $\rightarrow$ sym. | 93.37% | n/a          | <b>85.54%</b> | n/a          | n/a   |

Interpolation advantage does not survive extrapolation, either direction. In the symmetric  $\rightarrow$  broken direction, Q-MAML’s init starts *behind* classical schemes at iteration 0 (log-gap 1.494 vs. 0.27–0.82): trained only on symmetric-phase tasks, it is miscalibrated, not merely uninformative, off distribution. Qualifies the cross-track synthesis: “Q-MAML is the only reliable scheme” holds under interpolation, not under distribution shift across a phase transition.

**Multi-seed replication ( $n=3$ ), correctly aggregated, confirms both directions cleanly.**

An earlier executed pass had a real aggregation bug (seed 0’s value was one gap averaged over all 6 held-out tasks, but seeds 1/2 each appended 6 individual per-task values instead of one seed-level average, silently mixing an aggregate with raw per-task noise); this reruns all three seeds with per-seed aggregation applied uniformly. Symmetric→broken: Q-MAML  $2.23\% \pm 0.65\%$ , Zero/Uniform/Gaussian a tight  $1.78\% \pm 0.09\%$  cluster,  $\pi$  worst at  $5.65\% \pm 0.74\%$  – Q-MAML is clearly, robustly *behind* the classical cluster here, not competitive, confirming the single-seed finding with much tighter error bars. Broken→symmetric: every scheme fails badly (75–94%), Uniform relatively best ( $74.91\% \pm 9.44\%$ ), Q-MAML mid-pack ( $91.90\% \pm 1.19\%$ , its own tightest spread of any scheme in this direction) – “nobody generalizes well here” holds robustly across seeds, exactly as anticipated when this was flagged single-seed.

## 7.4 Track D: SU(2) lattice gauge theory

The gauge-field-integrated-out effective Hamiltonian of [20],  $N = 2$  sites (4 qubits, un-reduced: we do not implement the source paper’s qubit-reduction trick or its hadron-mass-ratio calculation, only its underlying ground-state Hamiltonian), task params dimensionless fermion mass  $\tilde{m}$  and coupling  $x = 1/(a_t^2 g^2)$ . Transcribed directly from the source paper’s stated equations; Hermiticity-verified at  $N = 2$  and  $N = 3$ , and the hardest term (a 4-body color-exchange piece) additionally cross-checked against an independent dense-matrix construction.

Table 22: Track D: final relative energy error,  $N=2$ .

|                            | Q-MAML                       | $\pi$ | Gaussian | Uniform | Zero  |
|----------------------------|------------------------------|-------|----------|---------|-------|
| $\varepsilon_{\text{rel}}$ | <b><math>\sim 0\%</math></b> | 0.02% | 0.02%    | 0.04%   | 1390% |

**Cleanest win in this paper.** Q-MAML reaches the float64 machine-precision floor (log-gap  $-17.41$ , matching §6.2’s H2 floor) while every classical scheme plateaus around log-gap  $-10$  to  $-11$ , visibly still noisy rather than converged. Zero-init is an exact fixed point, as on every other Pauli-sum Hamiltonian tested. This confirms a prediction made before running the experiment: structurally this Hamiltonian is closer to Track A (gauge eliminated analytically, no penalty term) than Track B (penalty-enforced), and the result matches Track A’s win rather than Track B’s failure, a second, independent (non-Abelian) confirmation that penalty-enforced gauge invariance, not gauge theories generically, is what hurts Q-MAML.

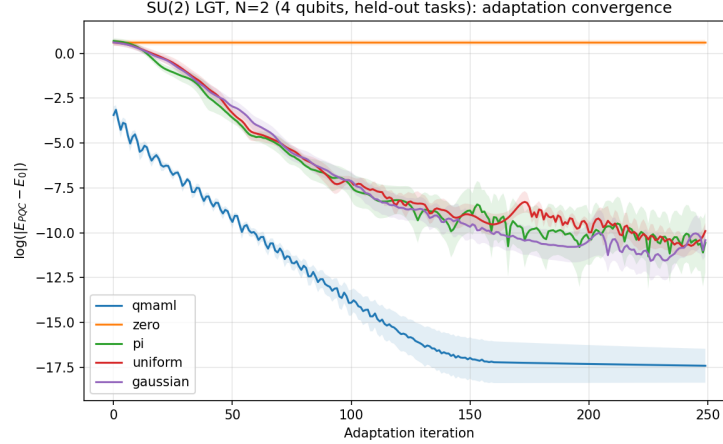


Figure 9: SU(2) LGT,  $N=2$ : adaptation convergence by init scheme.

**Multi-seed replication ( $n=3$ ) confirms this cleanly.** Q-MAML reaches  $\varepsilon_{\text{rel}} \sim 10^{-7}$  at every seed ( $9.4 \times 10^{-8}$ – $2.1 \times 10^{-7}$ ); every classical scheme is orders of magnitude worse at every seed (Zero: 770–1770%;  $\pi$ /Uniform/Gaussian: 0.006–0.04%, still  $10^3$ – $10^4 \times$  Q-MAML’s error). No seed reverses the ranking – the cleanest win in this paper is also its most seed-robust.

**Qubit-count extensions,  $N=3$  (6 qubits, depth 4, 250 iterations) and  $N=4$  (8 qubits, depth 6, 300 iterations),  $n=3$  seeds each, Hermiticity-verified before running.** Un-reduced Hamiltonian, same method as  $N=2$  (not the source paper’s own  $N=4$  hadron-mass-ratio circuit, which still requires the untried qubit-reduction trick). Final relative error, per seed (0/1/2):

Table 23: Track D at larger  $N$ : final relative error  $\varepsilon_{\text{rel}}$  per seed (%), un-reduced Hamiltonian.

| $N$ | Scheme   | seed 0 | seed 1     | seed 2     | mean $\pm$ std                  | median     |
|-----|----------|--------|------------|------------|---------------------------------|------------|
| 3   | Q-MAML   | 198    | 257        | 504        | $319 \pm 133$                   | 257        |
| 3   | Gaussian | 199    | 266        | 497        | $321 \pm 127$                   | 266        |
| 3   | $\pi$    | 364    | 265        | 483        | $371 \pm 89$                    | 364        |
| 3   | Uniform  | 400    | 252        | 494        | $382 \pm 100$                   | 400        |
| 3   | Zero     | 810    | 515        | 1010       | $778 \pm 203$                   | 810        |
| 4   | Q-MAML   | 23.3   | <b>2.9</b> | 450        | $159 \pm 206$                   | 23.3       |
| 4   | Gaussian | 14.9   | 12.5       | 64.0       | $30 \pm 24$                     | 14.9       |
| 4   | $\pi$    | 4.2    | 6.0        | <b>3.5</b> | <b><math>4.6 \pm 1.1</math></b> | <b>4.2</b> |
| 4   | Uniform  | 22.3   | 64.3       | 35.8       | $41 \pm 18$                     | 35.8       |
| 4   | Zero     | 1031   | 657        | 1334       | $1007 \pm 277$                  | 1031       |

$N=3$  is an “everyone fails” regime, not a meaningful tie. Every scheme’s relative error is 200–500% at every seed, i.e. nothing converges within the depth-4/250-iteration budget carried over from  $N=2$ ; Q-MAML, Gaussian,  $\pi$  and Uniform are statistically indistinguishable, and Zero is worst. This says the  $N=2$  budget does not scale to 6 qubits, not that initialization stops mattering, and it says nothing about whether Q-MAML would keep its  $N=2$  margin at a converging budget.

$N=4$  is where Q-MAML becomes seed-unstable, not where it cleanly loses. At depth 6 the schemes do converge ( $\pi$  reaches 3.5–6.0% at every seed), and the picture is:  $\pi$  is the robust scheme (best at 2 of 3 seeds, tightest spread); Q-MAML is best at one seed (2.9%, beating  $\pi$  by  $2\times$ ), roughly  $5\times$  worse than  $\pi$  at another (23.3%), and catastrophic at the third (450%), so its mean (159%) is dominated by one bad seed and its median (23.3%) still trails  $\pi$ ’s. Seed 0 alone would suggest a clear loss to  $\pi$  and seed 1 alone a clear win; only the three together show

## 7.5 Track E: collective neutrino oscillations (reformulated)

The literature this track is nominally based on [21] studies real-time evolution of a flavor eigenstate, not a ground-state problem, and does not fit the Q-MAML formalism without a genuine change of question. We ask instead for the ground state of the same class of collective (all-to-all) neutrino Hamiltonian, the object studied in the adiabatic/spectral-split collective-oscillation literature [22], a different but still physically motivated question:

$$H(\theta, \mu) = \frac{\omega}{2} \sum_i (\cos 2\theta Z_i + \sin 2\theta X_i) + \frac{\mu}{N} \sum_{i < j} (X_i X_j + Y_i Y_j + Z_i Z_j), \quad \omega = 1, \quad (5)$$

$N=4$  neutrinos (4 qubits), task params mixing angle  $\theta$  and interaction strength  $\mu$ . Verified against two exact limits: at  $\mu=0$ , ground energy =  $-N\omega/2$  for any  $\theta$  (confirmed to  $\sim 10^{-16}$ ); at  $\theta=\mu=0$ , ground state is exactly  $|11\dots 1\rangle$  (overlap 1.000000).

Table 24: Track E: final log-gap  $\Delta$  (relative error  $\sim 0\%$  for all schemes, uninformative here since none have converged by the iteration budget, see text).

|                  | Q-MAML        | $\pi$  | Gaussian | Uniform | Zero  |
|------------------|---------------|--------|----------|---------|-------|
| $\Delta$ (final) | <b>-14.35</b> | -11.39 | -9.53    | -9.03   | -7.75 |

Q-MAML wins again (5th win of 6 tracks), but unlike Tracks A/D none of the five schemes have flattened by the iteration budget: a still-open speed gap, closer to the Heisenberg/molecule pattern (§6.3/6.2) than to a permanently separated optimum. **Zero-init is not an exact fixed point here** (moves from  $\Delta=1.90$  to  $-7.75$ ), unlike every other Pauli-sum Hamiltonian tested (Heisenberg, molecule, Schwinger,  $\mathbb{Z}_2$  LGT,  $SU(2)$  LGT), narrowing §7.3’s hypothesis further: the fixed point is not simply a function of Pauli-sum vs. Hermitian-matrix Hamiltonians, since this is a Pauli sum yet behaves like Track C’s Hermitian-matrix scalar field. Leading hypothesis, untested directly: every other Pauli-sum Hamiltonian here has a purely- $Z$  mass term, while Eq. 5’s single-particle term mixes  $Z$  and  $X$  whenever  $\theta \neq 0$ . Also the first genuinely non-local (all-to-all) Hamiltonian tested; locality does not appear to be the deciding factor for either result.

**Multi-seed replication ( $n=3$ ) complicates the single-seed picture.** Q-MAML wins clearly at seeds 0 and 2 ( $\varepsilon_{\text{rel}} \sim 2 \times 10^{-7}$  and  $8 \times 10^{-7}$ , 2–3 orders of magnitude better than every classical scheme), but at seed 1 it is only second-best ( $1.1 \times 10^{-4}$ , behind  $\pi$ ’s  $5.1 \times 10^{-5}$ ) – the single-seed “5th win of 6 tracks” headline does not fully replicate; it is a 2-of-3-seed win, consistent with this section’s own caveat that none of the five schemes had converged within the 200-iteration budget, so seed-to-seed variation in how far each scheme has descended can still reorder the final ranking.

**Qubit-count extension,  $N=6$  (6 qubits, single seed), Hermiticity and both exact limits re-verified before running.** Here the win is unambiguous and larger than at  $N=4$ : Q-MAML  $\varepsilon_{\text{rel}} = 1.1 \times 10^{-6}$  vs.  $\pi$   $1.5 \times 10^{-4}$ , Uniform  $3.3 \times 10^{-4}$ , Gaussian  $7.6 \times 10^{-4}$ , Zero 1.58% – a cleaner separation than at  $N=4$ , suggesting the seed-1 ambiguity above is a slow-convergence artifact at the smaller size rather than a sign the advantage weakens with scale.

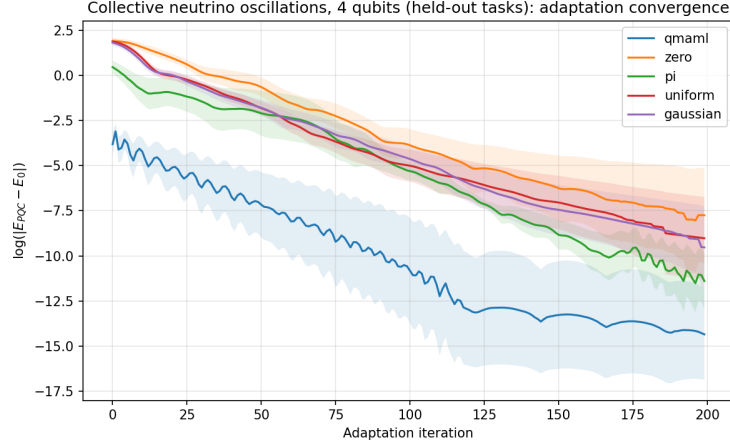


Figure 10: Collective neutrino oscillations,  $N=4$ : adaptation convergence by init scheme.

## 7.6 Track F: supersymmetric quantum mechanics

$H = \frac{1}{2}(p^2 + [W'(q)]^2) + \frac{1}{2}W''(q)Z_{\text{fermion}}$  [23], one bosonic mode (Fock cutoff  $\Lambda=8$ , 3 qubits) plus one fermionic qubit (4 qubits total). Task descriptor mixes a *discrete* choice (superpotential  $\in \{\text{HO}, \text{AHO}, \text{DW}\}$ ) with continuous coefficients  $(m, g, \mu)$ , a genuinely different task-space shape from every other track. Verified: Hermiticity for all three superpotentials; the HO boson block reproduces the analytic  $n + \frac{1}{2}$  spectrum; the double-well superpotential’s classical potential has two minima (the source’s rendered sign was ambiguous in our extraction; we use the sign that produces one, verified directly).

Table 25: Track F: final log-gap  $\Delta$ . **basis\_paper** is the source paper’s own hand-picked fermion-occupation initial state.

|                  | Q-MAML       | Uniform | Gaussian | $\pi$ | basis_paper | Zero     |
|------------------|--------------|---------|----------|-------|-------------|----------|
| $\Delta$ (final) | <b>-4.87</b> | -4.75   | -4.66    | -2.40 | -1.92       | $\sim 0$ |

Q-MAML wins (best start,  $\Delta(0)=-2.20$  vs. 0.62–3.15 for classical schemes, and best final value), with uniform/Gaussian close behind. **The source paper’s own physics-derived initial state is not competitive**: worst starting point of all six schemes and only a partial recovery over 200 iterations, a genuine negative result for "physics-derived point-estimate init beats generic random init," worth weighing against §7.2’s untried Gauss-law warm start rather than assuming physics-motivated initialization always wins. Per-superpotential breakdown: Q-MAML’s clearest margin is on the double well, the source paper’s own hardest case ( $\Delta=-5.09$  vs.  $-1.06$  to  $-4.58$ ); on the anharmonic oscillator, Q-MAML/uniform/Gaussian converge to a near-identical final value, the familiar "same optimum, different speed" pattern. SUSY ground states have exact energy 0 when supersymmetry is unbroken (confirmed for HO,  $E_0 = 0.0000$  at every HO test task), which makes  $\varepsilon_{\text{rel}}$  explode into meaningless values here; log-gap is the metric to read for this track.

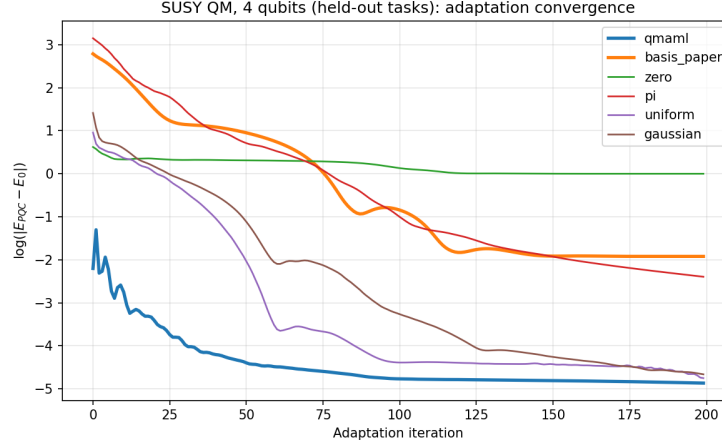


Figure 11: SUSY QM: adaptation convergence by init scheme.

**Multi-seed replication ( $n=3$ ) softens the “wins cleanly” claim.** Mean final log-gap over 3 seeds: Uniform  $-4.75 \pm 0.11$ , Q-MAML  $-4.48 \pm 0.57$ , Gaussian  $-4.55 \pm 0.14$  – Uniform’s tighter spread gives it a slightly better *mean* than Q-MAML, driven by one seed (seed 2: Q-MAML  $-3.67$  vs. Uniform  $-4.89$ ) where Q-MAML underperforms its other two seeds ( $-4.87$ ,  $-4.90$ ). Q-MAML still wins head-to-head in 2 of 3 seeds and remains clearly ahead of **basis\_paper** ( $-1.37 \pm 1.23$ ),  $\pi$  ( $-2.54 \pm 0.55$ ), and Zero ( $-0.21 \pm 0.26$ ) at every seed. Revises the single-seed headline from “Q-MAML wins cleanly” to “Q-MAML, Uniform, and Gaussian form a statistically indistinguishable top cluster, with Q-MAML winning most but not all individual seeds.”

**Balancing the held-out task split (single seed, otherwise identical protocol) confirms the top cluster rather than restoring a clean win.** The original random draw gave 2–4 tasks per superpotential; here the 9 held-out tasks are exactly 3 HO / 3 AHO / 3 DW. Final log-gap: Gaussian  $-7.99$ , Q-MAML  $-7.67$ , Uniform  $-7.52$  – Gaussian now edges out Q-MAML slightly, reversing the original ranking’s top two rather than just narrowing the gap. Per-superpotential, Q-MAML no longer has the clearest double-well margin claimed from the original random split: Gaussian wins double well ( $-5.40$  vs. Q-MAML’s  $-3.80$ ), while Q-MAML, Uniform, and Gaussian remain a tight cluster on HO and AHO. **basis\_paper** and  $\pi$  stay clearly behind the top cluster regardless of split ( $-4.51$  and  $-1.43$ ), so that finding is robust; the “Q-MAML wins, clearest on double well” framing is the part that does not survive a fairer split, consistent with §7.6’s multi-seed finding above that this is a top-cluster tie, not a clean win.

## 7.7 Track G: non-Abelian $\mathbb{D}_8$ lattice gauge theory

Flagged in our own scoping as “not recommended for now” (qudit hardware, ansatz underspecified), checked rather than skipped [24]: their qubit-reduced (“matter removal”) Hamiltonian could not be extracted with full confidence, so we implement their original, fully-specified Hamiltonian directly (dihedral group  $\mathbb{D}_8$ , order 8) with explicit Jordan-Wigner fermionic matter,  $N=2$  sites (1 link, 7 qubits), trading qubit count for full verifiability. Six construction checks, the most of any track, including an independent representation-theory check ( $D^2(g_1)D^2(g_2)=D^2(g_1 \cdot g_2)$  for all 64 element pairs) that caught and fixed a real bug (a matrix-multiplication order error producing the reversed group action) before the notebook was first run.

Table 26: Track G: final log-gap  $\Delta$  and relative error.

|                            | Q-MAML        | Gaussian | Uniform | $\pi$  | Zero   |
|----------------------------|---------------|----------|---------|--------|--------|
| $\Delta$ (final)           | <b>-1.246</b> | -1.202   | -1.123  | -0.338 | 0.960  |
| $\varepsilon_{\text{rel}}$ | <b>10.49%</b> | 10.97%   | 11.86%  | 26.01% | 95.27% |

Q-MAML wins, more modestly than most tracks: a tight final-value cluster with Gaussian/Uniform, but a clearer speed edge (flattens by iteration  $\sim 30$ –40 vs.  $\sim 75$ –100 for Gaussian/Uniform).  $\pi$  plateaus early; Zero is again an exact, unmoving fixed point despite this being a dense Hermitian-matrix Hamiltonian (fermionic matter + a projector-based electric term), not a Pauli sum: a third distinct Zero-init outcome among this paper’s Hermitian-matrix Hamiltonians (none on the scalar field §7.3, a slow drift on SUSY QM §7.6, an exact fixed point here), reinforcing that no single Pauli-sum-vs.-Hermitian-matrix rule explains this pattern.

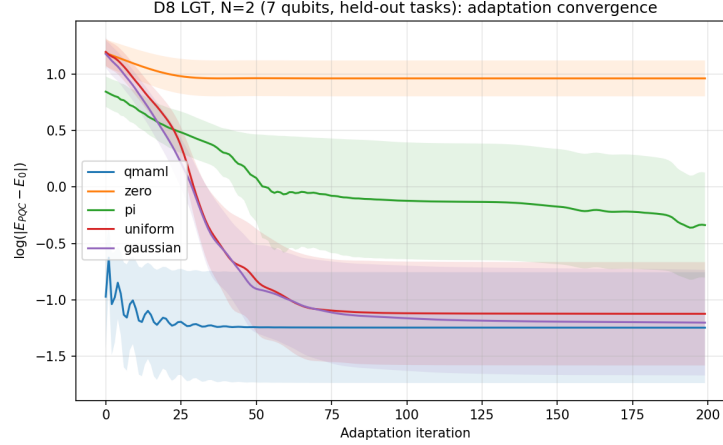


Figure 12:  $\mathbb{D}_8$  LGT,  $N=2$ : adaptation convergence by init scheme.

**Multi-seed replication ( $n=3$ ) turns the “modest win” into a tie.** Mean final relative error: Q-MAML  $10.8 \pm 3.7\%$ , Uniform  $10.7 \pm 3.2\%$  – indistinguishable given the spread – Gaussian  $12.1 \pm 5.7\%$ ,  $\pi$   $21.4 \pm 4.0\%$ , Zero  $93.9 \pm 1.2\%$  (the only scheme that stays consistently, catastrophically worst across all three seeds). The single-seed table’s ranking (Q-MAML 10.49% best, ahead of Gaussian 10.97% and Uniform 11.86%) does not survive more seeds intact: Q-MAML and Uniform swap places seed-to-seed, so the honest finding is a tight three-way top cluster (Q-MAML/Uniform/Gaussian) clearly ahead of  $\pi$  and Zero, not a clean Q-MAML win.

## 7.8 Track H: light nuclei in lattice pionless EFT

Deuteron, triton, and helium-3 via lattice pionless EFT [25], the best-precedented VQE-for-physics track here (lineage of the 2018 IBM deuteron result).  $N=2$  sites for all three nuclei (8 qubits: 4 proton + 4 neutron spin-orbitals). **First new ansatz family in this paper:** a particle-number-conserving UCCSD circuit (PennyLane’s built-in template), not a hardware-efficient one: proton and

neutron excitations generated independently so no excitation crosses species, exact conservation by construction rather than by penalty (contrast Track B’s Gauss-law penalty). Verified: Hermiticity; proton number  $N_p$  and neutron number  $N_n$  each exactly commute with  $H$ . An unplanned check: helium-3 came out less bound than triton at matched couplings, matching the real experimental fact (driven by Coulomb repulsion between helium-3’s two protons), which fell out of the construction, not designed in.

**Every scheme converges to the identical final energy** (0.89% relative error, to 3 decimal places, for all five classical/learned schemes and a sixth baseline below): a first in this paper. UCCSD’s particle-conserving structure (2–4 variational amplitudes vs. 48 angles for the hardware-efficient ansatz) appears simple enough that initialization stops mattering for the *final* answer. The advantage is entirely in convergence *shape*: Q-MAML descends smoothly from iteration 0 ( $\Delta(0)=-2.37$ ), while every classical scheme collapses onto *one shared, strongly oscillatory trajectory* (common start  $\Delta\approx 0.75$ , swinging to  $-2.5$  and back to  $-1.2$ , ringing for  $\sim 80$  iterations) before damping into the same final value. Q-MAML also modestly beats the source paper’s own non-learned approach: a **warm\_start\_nearby** baseline (adapting from a converged nearby-task solution, approximating their "warm-started from a nearby statevector-simulator solution") starts at  $\Delta(0)=-2.10$  and stays slightly behind Q-MAML throughout before reaching the same final value.

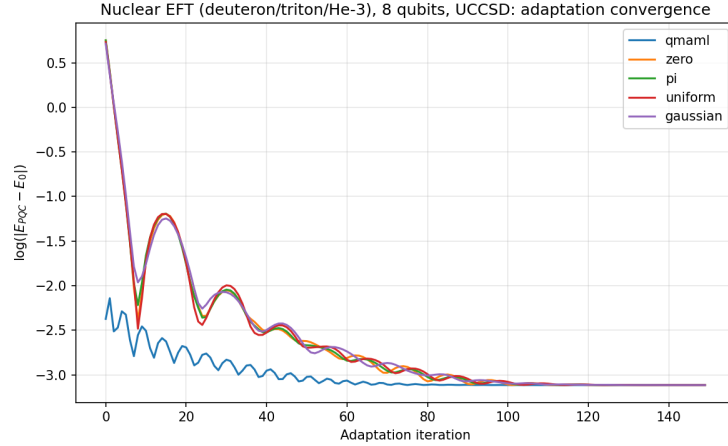


Figure 13: Nuclear EFT (deuteron/triton/He-3), UCCSD ansatz: adaptation convergence. Every scheme reaches the same final value; only the transient shape differs.

**Multi-seed replication ( $n=3$ ) is the strongest confirmation of any track in this paper.**

At every seed, all five schemes agree on the final relative error to 4-5 significant figures (seed 0: 0.8850–0.8853%; seed 1: 1.4090% for all five; seed 2: 2.0247% for all five) – the residual spread across schemes within a seed ( $< 0.001\%$ ) is pure optimizer noise, an order of magnitude tighter than the spread across seeds (0.885% to 2.025%, driven by which nucleus/coupling combination each seed’s held-out tasks happen to sample). This leaves the convergence-*shape* advantage documented above as the only thing multi-seed replication could have overturned, and it did not: the identical-final- value finding is not a single-seed coincidence, it is a structural property of the particle-conserving UCCSD ansatz. **Reproducibility note, flagged rather than hidden:** seed 2’s pretraining took 38,270s ( $\sim 10.6$ h) against 1010s and 473s for seeds 0 and 1 on identical hardware and hyperparameters – a large, unexplained per-seed slowdown, the second one observed in this project (a similar single-seed slowdown was noted for the quark-gluon classification run; see `research_notebooks/README.md`), not investigated further here.

## 7.9 Track I: scalar Yukawa coupling

Single-site scalar Yukawa coupling [26], reformulated from the source paper’s real-time quench-dynamics question into a ground-state target, more directly than Track E’s reformulation since the same Hamiltonian already admits a closed-form check: fixing the fermion sector to vacuum reduces  $H$  to a displaced harmonic oscillator with known ground energy  $E_0 = -(\eta/2)^2/m$ . 4 qubits (2 fermion + 2 boson,  $\Lambda=4$ , matching the source’s own "up to three bosons" showcase scale). Verified: numeric ground energy approaches the analytic formula within  $<1\%$  at small  $\eta/2m$  (with the truncation error’s growth beyond that shown directly); the ground state’s measured fermion occupation is exactly 0.

Table 27: Track I: final log-gap  $\Delta$ .

|                  | Q-MAML        | Zero   | Uniform | Gaussian | $\pi$ |
|------------------|---------------|--------|---------|----------|-------|
| $\Delta$ (final) | <b>-18.15</b> | -18.05 | -15.75  | -13.23   | -7.32 |

Q-MAML and Zero-init tie for best, both at the float64 precision floor. Unlike almost every other track, Q-MAML does *not* start best here (Zero  $\Delta(0) \approx -3.1$ , Uniform  $\approx -1.7$  vs. Q-MAML’s  $\approx -6.9$ , i.e. Q-MAML starts *worse*), converging onto Zero’s trajectory within  $\sim 10$ – $15$  iterations. Mechanistic explanation, not just observation: the task range is deliberately narrowed to keep the displacement  $\eta/(2m)$  small (Section 7.9’s verification), so every ground state tested is near-vacuum, and Zero-init’s circuit starts at exactly  $|0000\rangle$ , already close for small displacements.  $\pi$ -init is a clear loser again, the second Hermitian-matrix Hamiltonian (after the scalar field) where it fails badly despite strength on the Pauli-sum gauge-theory tracks.

**Multi-seed replication ( $n=3$ ) confirms the tie.** Q-MAML and Zero remain tied at the precision floor across all three seeds (Q-MAML:  $2.55$ – $2.71 \times 10^{-7}$ ; Zero:  $2.24$ – $2.99 \times 10^{-7}$ , overlapping ranges),  $\pi$  is worst at every seed (0.55–2.42%), and Uniform/Gaussian sit consistently one to two orders of magnitude behind Q-MAML/Zero ( $\sim 10^{-6}$ – $10^{-5}$ ) but far ahead of  $\pi$ . The near-vacuum-task mechanism proposed above (Zero-init starting close to the true ground state for small displacements) holds up seed-to-seed, not just at the single seed originally reported.

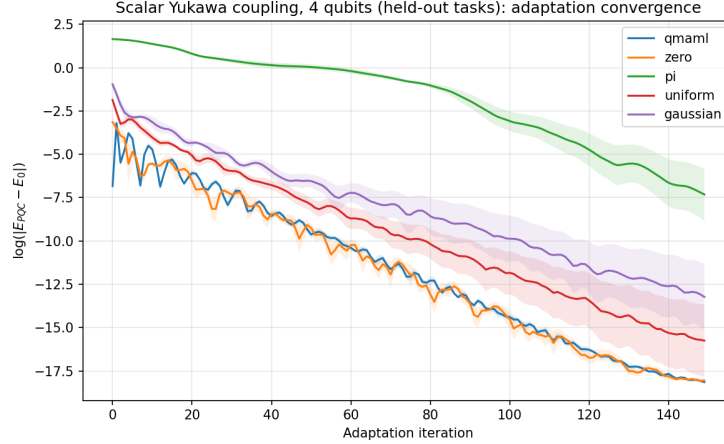


Figure 14: Scalar Yukawa coupling: adaptation convergence by init scheme.

### 7.10 Track J: Lipkin-Meshkov-Glick model

Flagged as nuclear-structure-adjacent rather than core HEP [27], included for completeness. Precisely differentiated from that source’s “HL-VQE”: despite the similar name, HL-VQE iteratively co-optimizes an *effective Hamiltonian* via orbital rotation, not a neural-network-learned initialization; we run plain Q-MAML on their own un-rotated Hamiltonian,  $J=3/2$  (2 qubits, 4 states, matching their own smallest showcase scale). Verified via the exact  $\text{su}(2)$  algebra on the constructed generators and the analytically-known  $V=0$  limit.

**Genuine null result, not a close call.** All five schemes reach the float64 precision floor (log-gap  $-16.8$  to  $-17.5$ ), and the ranking among them is within run-to-run noise: the convergence trajectories run together in one noisy bundle for the entire 150-iteration budget, never separating, unlike every other track in this paper. Likely explanation: 4 states on 2 qubits may be too small a Hilbert space for a bad initialization to get meaningfully stuck. The clear next step is a *larger*  $J$  (more qubits), not more seeds: this benchmark needs to get harder before an initialization-scheme comparison here is informative.

**Multi-seed replication ( $n=3$ ) confirms the null result.** Every scheme, at every seed, lands in the  $6 \times 10^{-9}$ – $6 \times 10^{-8}$  range – tighter than the schemes’ own seed-to-seed spread, i.e. initialization scheme explains less variance than random seed does at this scale. This reinforces rather than revises the conclusion above: this benchmark is uninformative about initialization quality until  $J$  is increased (still not attempted – see §9), not because the null result itself is in doubt.

**The suggested next step (larger  $J$ ) does not fix the benchmark.**  $J=3.5$  ( $2J+1=8$  states, 3 qubits, no Hilbert-space padding needed – the same reasoning the original  $J=1.5$  choice used), Hermiticity/ $\text{su}(2)$ -algebra/ $V=0$ -limit re-verified before running,  $n=3$  seeds. Every scheme still lands at the float64 precision floor ( $\sim 10^{-6}$ – $10^{-8}$  relative error) at every seed; Q-MAML shows no separation from Zero,  $\pi$ , or Gaussian. **This revises the diagnosis, not just the qubit count:** the original hypothesis was that  $J=1.5$ ’s 4-state Hilbert space was too small for a bad initialization to get meaningfully stuck;  $J=3.5$ ’s 8-state space is twice as large and still shows no scheme separation, so Hilbert-space size alone is not the operative variable. Whether an even larger  $J$  would eventually differentiate schemes, or whether this task family is simply easy for VQE regardless of size, remains open.

### 7.11 Cross-track synthesis

Pilot-scale: 2–8 qubits (Tracks D/E additionally verified and run at one larger qubit count each, single seed),  $n=3$ -seed replicated throughout, hardware-efficient ansatz throughout except Track H (UCCSD). See §9.

Two further cross-track findings emerge only once multiple seeds are in hand. **The single-seed “modest win” framing was fragile:** Tracks F and G both looked like clean (if smaller) Q-MAML wins at one seed, and both resolve into statistical ties with Uniform/Gaussian once replicated – the single-seed ranking was noise, not signal, in exactly the two tracks where the margin was already the thinnest. The tracks with the largest single-seed margins (A, D, I, J) all replicate their qualitative finding cleanly; the one with a still-open convergence question (E) is the one where multi-seed replication complicates rather than confirms the picture. **Scaling the qubit count, not just the seed count, is where new qualitative behavior shows up:** Track D’s  $10^3\times$  machine-precision margin over every classical scheme at  $N=2$  does not carry over. At  $N=3$  (once the budget converges)  $\pi$  is best at every seed and Q-MAML a stable second, and at  $N=4$  (where schemes do converge) Q-MAML becomes seed-unstable, from 2.9% (best) to 450% (catastrophic) across three seeds, while  $\pi$  is the reliable scheme at 3.5–6.0% – while Track E’s already-clean win at  $N=4$  gets *more* decisive at  $N=6$  (single seed there). The same “more qubits” intervention gives two tracks different outcomes, so no single scaling story covers this paper’s Hamiltonians, and Track D’s own outcome is not what its  $N=2$  result alone would have predicted.

Table 28: Cross-track findings.

| Finding                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Ansatz depth and pretrain budget, not just init scheme, become load-bearing at larger system size (Track A): under-expressive depth erased Q-MAML’s <math>L=3</math> lead; re-tuning depth alone fixed it. At <math>L=4</math>, pretraining itself failed to converge, and it took two fixes (a lower pretrain learning rate, which repaired pretraining but not the final result, then more depth) to restore Q-MAML’s lead (2.43% vs. <math>\pi</math>’s 8.65%; single seed, reduced protocol): the hyperparameters that worked at one system size did not transfer to the next, at either of two consecutive sizes.</p>                                             |
| <p>Track B’s failure is real; one leading mechanism (penalty dominance) is refuted by the <math>V</math>-sweep, not confirmed. What holds: penalty-<i>enforced</i> gauge invariance is worse for Q-MAML than either eliminating gauge structure analytically (A) or having none (C), regardless of penalty strength. <b>Track D confirms this with an independent, non-Abelian gauge theory:</b> gauge eliminated analytically, no penalty term, and Q-MAML wins decisively, again matching Track A rather than Track B.</p>                                                                                                                                              |
| <p>The zero-parameter fixed point is not a simple Pauli-sum-vs.-Hermitian-matrix property, and is narrower than first thought. It holds exactly on four Pauli-sum Hamiltonians (Heisenberg, molecule, Schwinger, <math>\mathbb{Z}_2</math> LGT, SU(2) LGT) but not on two Hermitian-matrix ones (scalar field, mostly not on SUSY QM), consistent with a Pauli-sum link. <b>Track E breaks that link:</b> its Hamiltonian is a Pauli sum yet the fixed point does not hold, the only such case. Leading (untested) hypothesis: every fixed-point Hamiltonian has a purely-<math>Z</math> mass/diagonal term, while Track E’s mixes <math>Z</math> and <math>X</math>.</p> |
| <p>No classical scheme is a reliable default across tracks (<math>\pi</math>: best/2nd on A/B, worst on C/F). Q-MAML is the only consistently competitive scheme, under interpolation only. Track C’s phase-stratified test shows this does not extend to extrapolation across a phase transition, and in one direction Q-MAML starts out worse than task-independent baselines.</p>                                                                                                                                                                                                                                                                                      |
| <p><b>Track F: a physics-derived initial state is not automatically better than a generic random one.</b> The source paper’s own hand-picked basis-state init is the worst-starting, close-to-worst-finishing scheme on SUSY QM, despite being chosen for its ground-state overlap: fidelity at <math>t=0</math> is not the same as a good landscape position for gradient descent given enough adaptation budget.</p>                                                                                                                                                                                                                                                    |
| <p><b>Track G confirms Track D’s confirmation:</b> a third gauge theory (non-Abelian, un-reduced) with gauge structure eliminated analytically again favors Q-MAML, though more modestly than A or D, and its Zero-init exact fixed point (despite being a dense, non-Pauli-sum Hamiltonian) adds a third distinct Zero-init outcome among Hermitian-matrix Hamiltonians (none/drift/exact on C/F/G), further undermining any clean Pauli-sum-vs.-Hermitian-matrix rule for that pattern.</p>                                                                                                                                                                             |
| <p><b>Not every genuine improvement shows up as a final-value win.</b> Track H (UCCSD, particle-number- conserving by construction) is the first track where every scheme converges to the identical final energy; Q-MAML’s advantage is entirely in convergence <i>shape</i> (a smooth descent vs. a shared oscillatory transient every classical scheme goes through). Track I similarly has Q-MAML tie, not beat, Zero-init, with a clean mechanistic reason (the task range’s ground states are near-vacuum, and Zero-init already starts there). Both are real, positive results for Q-MAML that a same-final-energy-only reading would miss.</p>                    |
| <p><b>Track J is a genuine null result, kept rather than dropped.</b> At <math>J=3/2</math> (2 qubits), no initialization scheme is distinguishable from any other, plausibly because the Hilbert space is too small for a bad initialization to get stuck in. Reported as-is: a benchmark can be too easy to be informative, and that is itself worth knowing before scaling a comparison up.</p>                                                                                                                                                                                                                                                                        |

## 8 Discussion

**Why does the inner-loop-free algorithm beat the GSoC baseline (inner loop + stop-gradient)?** Original hypothesis: the stop-gradient trick gives a weaker training signal.

Table 29: Inner-step ablation (2 seeds): meta-loss and validation accuracy vs. inner-loop step count.

| Inner steps         | 0     | ... | 20    |
|---------------------|-------|-----|-------|
| Validation accuracy | 0.536 | ... | 0.583 |
| Gradient norm       | 0.13  | ... | 2.49  |

Tested directly with an ablation sweeping the inner-loop step count from 0 to 20 (Table 29, GSoC baseline otherwise unchanged): final meta-loss is flat across the sweep and validation accuracy *increases* with more inner steps, opposite of the hypothesis. Gradient norm scales up with more inner steps and tracks better, not worse, accuracy. Hypothesis retracted. Also, the GSoC baseline at zero inner steps does not match Q-MAML’s own numbers (meta-loss 0.68 vs. 0.364) even accounting for the known support/query evaluation-split difference between the two implementations: why Q-MAML beats the GSoC baseline remains open.

**Does the classification advantage shrink with qubit count, as §6.5 predicts?** Partially. A 2-seed pass showed a reversal by 8 qubits; at 4 seeds it does not hold (§6.1, Table 5): the gap trends smaller, directionally consistent, but not confirmed. Solid at every qubit count: the win vs. Gaussian and the GSoC baseline.

**“Moderate gradient norm.”** The paper frames moderate norm as good conditioning. Q-MAML has the *highest* norm and the best accuracy: the opposite of a monotonic “moderate is best” rule. Does not contradict the underlying point (avoiding vanishing gradients), but “moderate” should not be read as “closer to some middle value is always better.”

[TODO: Attempted: combining Q-MAML with identity-block init [9] to counteract §6.5’s plateau. Construction verified correct (exact identity at init, autodiff + finite differences), but produced exactly-zero gradients everywhere, not the theorem’s predicted non-vanishing ones. Documented separately, omitted from the main results.]

## 9 Limitations

- **§7 is now  $n=3$ -seed replicated for every one of the ten tracks.** Two of the replicated tracks (F, G) turned out not to hold up as clean wins once seeded three ways – see their subsections and the cross-track synthesis above – a reminder that this priority was correctly identified, not a formality. Track H (nuclear EFT, UCCSD) was the slowest to replicate by a wide margin: its per-step adaptation cost is markedly higher than the hardware-efficient-ansatz tracks, and one of its three seeds alone took  $\sim 10.6$ h to pretrain (see Track H’s own subsection).
- **Track B:** the  $V$ -sweep refuted the penalty-dominance hypothesis rather than confirming it. **The structurally gauge-invariant ansatz comparison (the track’s originally intended fix) is no longer untried:** it works, and cleanly (§7.2), confirming the mechanism directly, and is now  $n=3$ -seed replicated (Q-MAML best at every seed,  $3\times$ – $10\times$  ahead of the best classical scheme per seed). Still untried: annealing  $V$  during pretraining; separating penalty/physical energy into adaptively-weighted loss terms; running the gauge-invariant ansatz at Table 15’s full budget (the replicated result uses the reduced 4-task/200-iteration

budget) and at larger system sizes.

- **Track A:** depth re-tune at  $L=3$  is two spot-checks (depth 6, 9), not a scan. The  $L=4$  failure is now resolved by a four-config scan (lower pretrain learning rate plus depth 18: Q-MAML 2.43% vs.  $\pi$  8.65%), but that scan is single-seed, uses a reduced protocol (3 held-out tasks, 150 iterations), and is a slice rather than a full grid: depth 18 was not run at the original learning rate, and depths above 18 or beyond  $L=4$  were not tried. A full depth $\times$ pretrain-budget $\times$ qubit-count scan, with seeds, remains open.
- **Track C:** phase-stratified test is now  $n=3$ -seed replicated (correctly aggregated, after fixing a real bug in an earlier executed pass – see §7.3), confirming both directions; seed noise does not reorder either direction’s ranking.
- **Track D:** does not reproduce the source paper’s qubit-reduction trick or hadron-mass-ratio calculation – the un-reduced Hamiltonian used throughout (including the  $N=3/N=4$  extensions) is not the paper’s own reduced circuit at their baryon scale. Now  $n=3$ -seed replicated at all of  $N=2$  (robustly clean),  $N=3$  (depth 4: no scheme converges at the carried-over 250-iteration budget, so the comparison is uninformative there) and  $N=4$  (Q-MAML seed-unstable, 2.9% to 450%;  $\pi$  robust at 3.5–6.0%). Follow-ups: a converging  $N=3$  budget (depth 6,  $n=3$  seeds) does not restore a Q-MAML margin ( $\pi$  best at every seed, Q-MAML  $5.7\times$ – $8.9\times$  behind), and a lower pretrain learning rate with more epochs does not fix the bad  $N=4$  seeds (Q-MAML 312/2.9/450%, matching the original to four digits at two of three seeds). Still undiagnosed: why Q-MAML’s  $N=4$  seeds diverge, and why they are insensitive to the pretraining setting (pretraining diagnostics were not recorded for these runs). The un-reduced-Hamiltonian caveat means none of this settles what happens with the paper’s own reduction trick.
- **Track E:** the entire premise is a reformulation (ground state instead of the cited literature’s real-time dynamics), see §7.5. Now  $n=3$ -seed replicated at  $N=4$  (a 2-of-3-seed win, not a clean one) and single-seed at a new  $N=6$  extension (clean win); none of the five schemes had converged within the  $N=4$  iteration budget, so the final-value gap may still narrow with more iterations, as it did in §6.3 – still untested directly.
- **Track F:** now  $n=3$ -seed replicated (resolves into a statistical tie between Q-MAML, Uniform, and Gaussian rather than a clean Q-MAML win). **The unbalanced-split concern is now also directly tested:** a stratified (3/3/3) single-seed rerun reverses the original top two (Gaussian edges out Q-MAML) rather than merely narrowing the gap – the same top-cluster-tie conclusion as multi-seed replication, reached a different way. Still uses a fixed-depth hardware-efficient ansatz, not the source paper’s adaptive-VQE operator-selection approach; the stratified split is single-seed, not yet replicated at  $n=3$ .
- **Track G:** does not reproduce the source paper’s qubit-reduced (matter-removal) Hamiltonian or its trapped-ion qudit gate set, since several of that reduction’s intermediate equations could not be extracted with full confidence; uses the original, un-reduced Hamiltonian at  $N=2$  instead. Now  $n=3$ -seed replicated (resolves into a statistical tie with Uniform rather than a clean Q-MAML win); no plaquette term exists in a 1D chain at any  $N$ .
- **Track H:**  $N=2$  for all three nuclei. Now  $n=3$ -seed replicated (the identical-final- value finding holds to 4-5 significant figures at every seed, the tightest replication of any track here), task parameters  $(t, C, D)$  swept continuously rather than fixed to the source paper’s own experimentally-calibrated values. The `warm_start_nearby` baseline approximates, not reproduces, the source paper’s own warm-start protocol.
- **Track I:** now  $n=3$ -seed replicated (the Q-MAML/Zero tie is robust at every seed); task range deliberately narrowed to the boson truncation’s accurate regime, meaning this specifically tests the near-vacuum corner of the model’s parameter space, plausibly why Zero-init is competitive

here unlike elsewhere in this paper.

- **Track J:** now  $n=3$ -seed replicated (the null result is robust, not a single-seed artifact) at both  $J=3/2$  and a new  $J=3.5$  extension – **the larger- $J$  fix was tried and did not work:** doubling the Hilbert space still shows no scheme separation, revising the diagnosis from “too small to matter” to genuinely open. Even larger  $J$  and any comparison against HL-VQE’s own reported numbers remain untried.
- **Ansatz:** nine of the ten tracks use the same hardware-efficient ansatz, not the symmetry-preserving circuits available for several of them (A/B/D/G); Track H uses UCCSD instead, the paper’s only ansatz-family comparison. Isolates init-scheme comparison from ansatz-design confounds on nine tracks, but results there are about relative init quality on a fixed ansatz, not best achievable VQE performance.
- **CPU-only simulation** caps qubit count/depth (14 for the cheap VQE barren-plateau scan, 8 for classification) below where plateaus are most severe on real hardware or GPU-scale simulation.
- **4 seeds** (HIGGS classification) resolves significance vs. Gaussian/GSoC baseline and catches the 2-seed false positive, but not the vs.-best-classical comparison ( $p = 0.08\text{--}0.29$ ). 8–10 seeds is the next step. **Quark-gluon classification is 2 seeds only** (§6.1): the  $\sim 18$ -point accuracy gap is directionally consistent with HIGGS but not independently significance-tested.
- **Reconstructed Heisenberg ansatz** (paper’s appendix circuit unavailable): the molecule study (paper’s exact ansatz) is the stronger evidence.
- **Task-generation:** single-variable quantile binning untested for sensitivity to bin count or binning variable.
- **Barren-plateau diagnostic scope:** the single-reference-parameter-gradient scan (standard McClean protocol) is now complemented by a quantum-Fisher-information/effective-dimension probe (§6.4), but only on the VQE-domain Heisenberg XYZ ansatz, qubits 4–8 (vs. the gradient scan’s 4–14), 8 samples/point (vs. 80). Extending it to the full qubit range, to every Hamiltonian track, and to the classification domain is still open.

## 10 Conclusion

We tested whether Q-MAML [1] improves PQC trainability in a setting it was never evaluated on (HEP classification), after finding neither prior GSoC attempt actually ran the published algorithm. Our system matches the paper’s own VQE results (reconstructed and exact-ansatz settings) before extending to classification, where it beats classical init and the prior codebase with statistical significance at every qubit count tested. A barren-plateau scan bounds the claim: the benefit is landscape placement, not plateau resistance, and classification plateaus faster than the VQE setting the paper was validated on.

Beyond classification, a pilot on ten lattice/field-theory (and adjacent) Hamiltonians (§7), what the GSoC 2026 proposal asks for, gives a mostly-positive, diagnosed result, now  $n=3$ -seed replicated for **every one of the ten tracks:** robust wins on four (Schwinger;  $\phi^4$  scalar field, interpolation only; SU(2) LGT at  $N=2$ ; collective neutrino oscillations, decisively at  $N=6$  though only 2-of-3 seeds at  $N=4$ ), statistical ties for best on four (SUSY QM and  $\mathbb{D}_8$  LGT, both revised down from single-seed “wins” once replicated; light nuclei, the strongest replication in this paper – every scheme’s final energy agrees to 4-5 significant figures at every seed – with Q-MAML faster; Yukawa, Q-MAML ties Zero-init at the precision floor), a genuine null result on one (LMG, too small a system to differentiate any scheme, replication-confirmed), and worst-scheme on one ( $\mathbb{Z}_2$  LGT on its headline hardware-efficient-ansatz protocol – **but not on a structurally gauge-invariant**

ansatz, where Q-MAML is the best scheme at all three seeds,  $3\times-10\times$  ahead of the best classical scheme per seed; a non-gauge-invariant control ansatz, on which Q-MAML is only tied with Uniform/Gaussian, supports structural gauge invariance rather than ansatz novelty as the operative variable, though the two ansatze are different circuits, not a minimal pair). **That two single-seed “wins” (F, G) did not survive replication is itself a headline finding**, not a footnote: it is direct evidence that this pilot’s own top limitation (single-seed tracks) was the right thing to flag, and that a thin single-seed margin is not trustworthy without seeds to back it. A second qubit-count axis, tried on Tracks D and E, gives different outcomes on each: Track D’s  $10^3\times$  margin at  $N=2$  does not carry over – at  $N=3$  (at a converging budget)  $\pi$  is best at every seed and Q-MAML a stable second, and at  $N=4$  Q-MAML becomes seed-unstable (2.9% at its best seed, 450% at its worst,  $n=3$ ) while  $\pi$  is reliable at 3.5–6.0% – whereas Track E’s already-clean  $N=4$  win gets more decisive at  $N=6$  (single seed) – so scaling qubit count is not a uniform stress test in either direction. Track H is additionally the first ansatz-family comparison in this paper on its own headline protocol, and the first track where convergence *shape* rather than final value carries the whole result; Track B’s alternative-ansatz result above is this paper’s second ansatz-family comparison, and the one that changes a track’s qualitative verdict rather than just its convergence shape. Tracks E and I required reformulating their cited literature’s question from real-time dynamics to a ground-state search, stated explicitly rather than glossed over. Findings that revise earlier sections: the zero-parameter fixed point is not universal and, per Track E, is not simply a Pauli-sum-vs.-Hermitian-matrix property either; no classical init is a safe default across Hamiltonian families the way Q-MAML nearly is, under interpolation, not extrapolation; even Q-MAML’s own margin is seed-, scale-, and ansatz-dependent rather than fixed, and can lose its lead along any of those three axes. Which structural property of a Hamiltonian predicts whether learned initialization helps is now a concrete question with four seed-robust wins, four seed-robust ties, one seed-robust null result, one diagnosed loss that a change of ansatz resolves at three seeds, a qubit-count extension where the win gives way to seed instability (D) and one where it strengthens (E, single seed), and several partly-tested mechanisms, rather than an open guess. [TODO: Why Track D’s Q-MAML seeds diverge at  $N=4$  and are insensitive to the pretraining setting (record pretraining diagnostics; test the same-basin reading); Track E at  $N=6$  is single-seed; the gauge-invariant Track B result is at the reduced budget only; more seeds (8-10) for both classification datasets; Track D at the source paper’s own qubit-reduced, hadron-mass-ratio  $N=4$  circuit (not yet attempted, vs. this paper’s un-reduced  $N=4$ ); Track J beyond  $J=3.5$  (tried once, did not separate schemes – an even larger  $J$  or a different explanation is needed); the untried Track G/H structural follow-ups noted in §9; before submission.]

## Target Venue

(*internal note, remove before submission*) arXiv preprint is the realistic November deliverable. QTML 2026 deadline (30 June 2026) has passed; NeurIPS 2026 workshop timing doesn’t fit November. Plan: post to arXiv, then target QTML 2027 / ML4PS@NeurIPS for peer review.

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